



Antimalarial, antiprotozoal & anthelmintic drugs

4 February 2026

Kodcharat Cheevapruk, PhD
Department of Pharmacology, Faculty of Pharmacy
Email: kodcharat.che@mahidol.ac.th

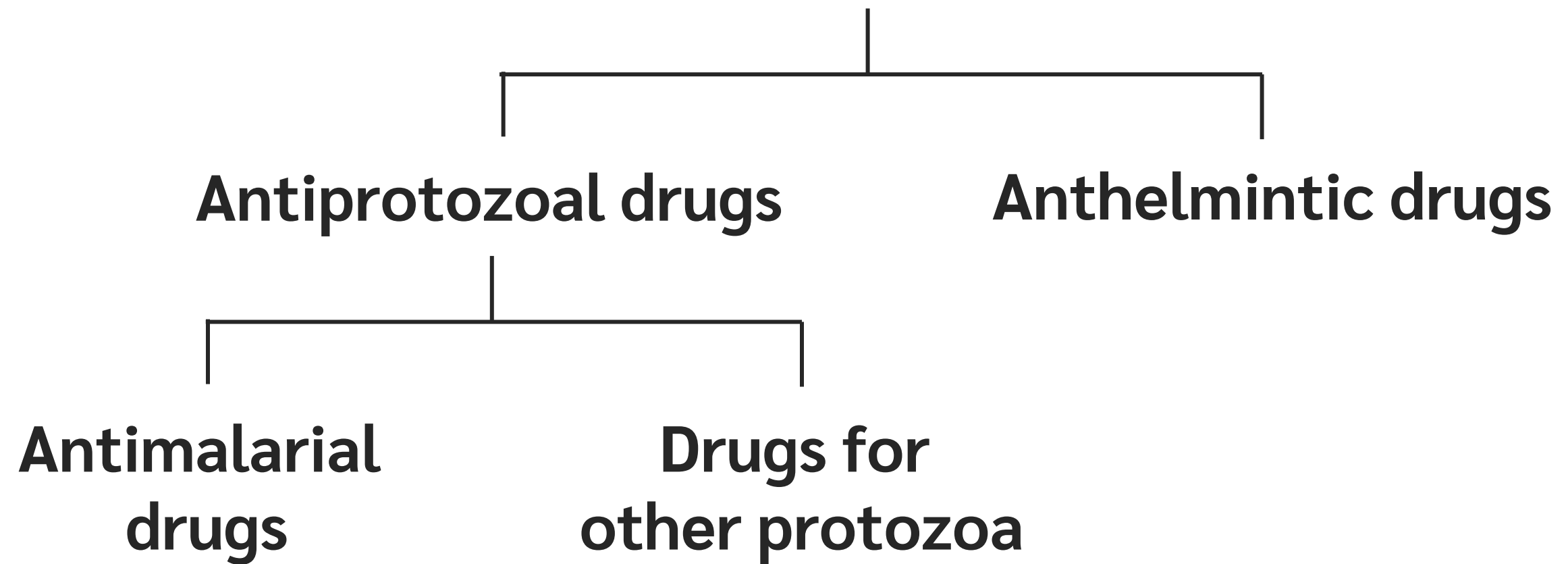


Objectives

After this class, students should be able to:

- explain the mechanisms of action of antimalarial, antiprotozoal & anthelmintic drugs.
- identify common adverse drug reactions & drug interactions.
- apply pharmacological principles to select appropriate therapy for specific parasitic infections.

Antiparasitic drugs

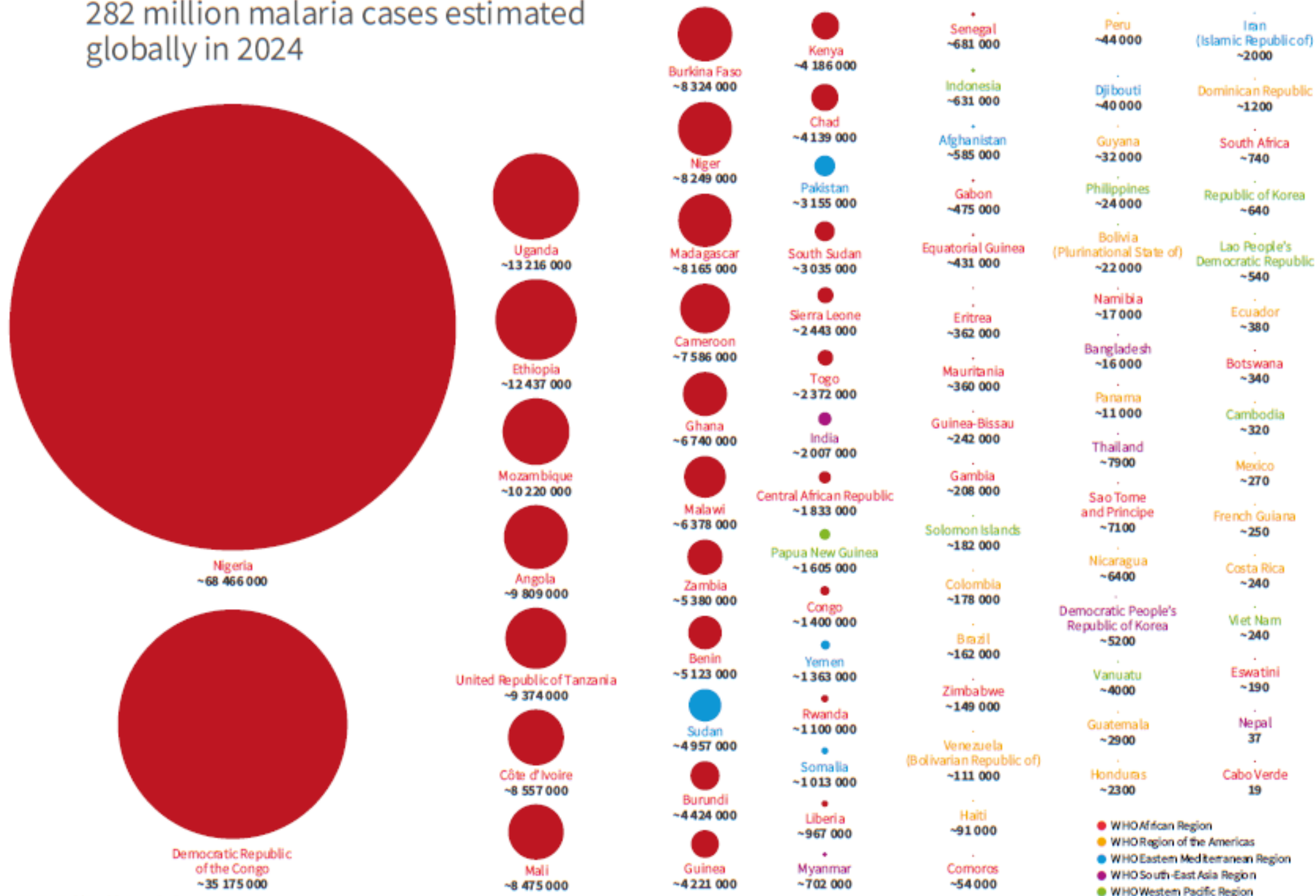




Antimalarial drugs

Estimated number of malaria cases per country & area in 2024

282 million malaria cases estimated globally in 2024



WHO: World Health Organization.



Species of plasmodium parasites

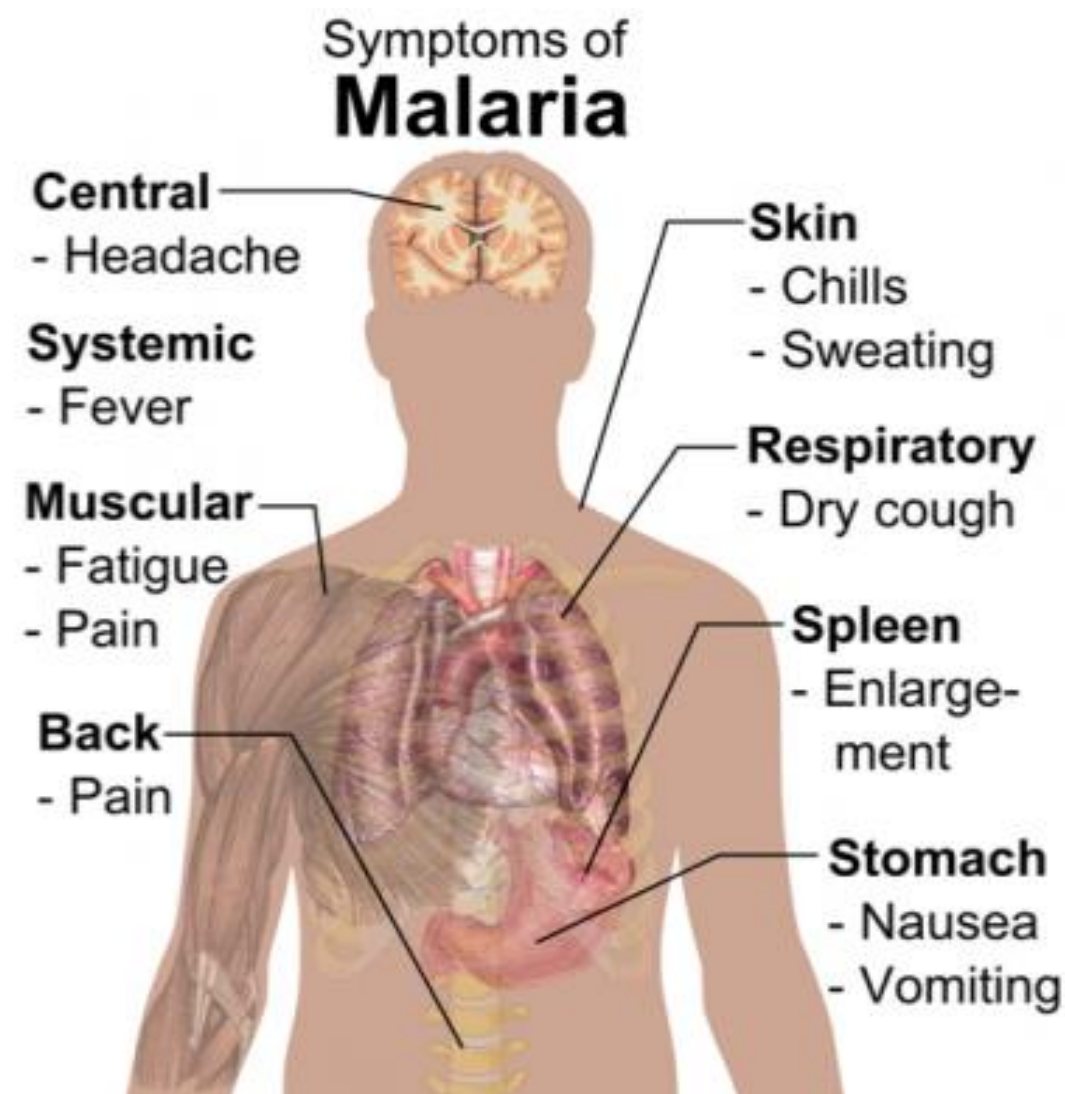
Five species of which are known to infect humans:

1. *Plasmodium falciparum*
incubation time 8-12 days
2. *Plasmodium vivax*
incubation time 10-15 days
3. *Plasmodium ovale*
incubation time 10-15 days
4. *Plasmodium malariae*
incubation time 30-40 days
5. *Plasmodium knowlesi*
incubation time 9-12 days

**P. falciparum* & *P. vivax* → most malarial infections worldwide (including Thailand).

**P. falciparum* → most severe disease → organ failure & death

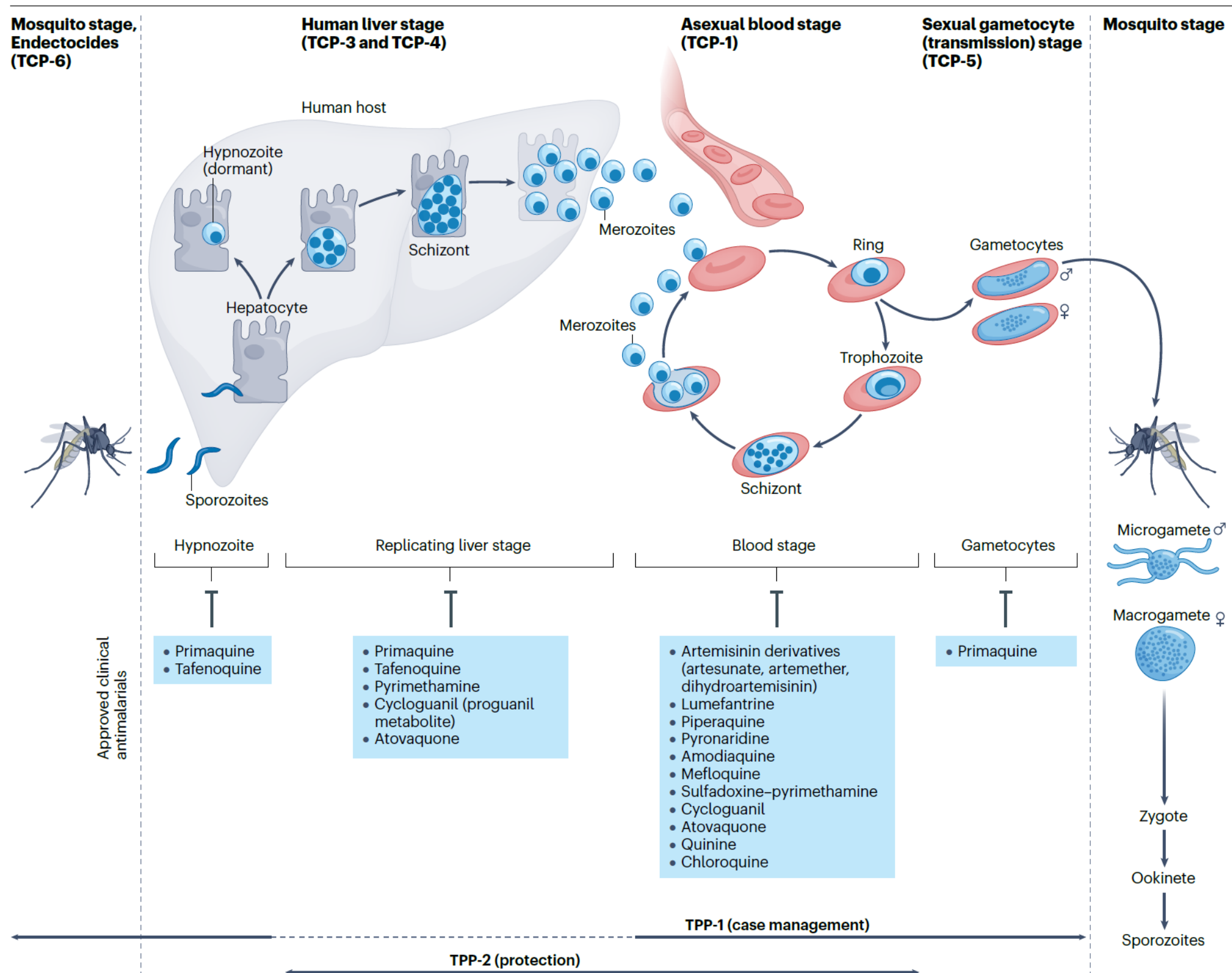
Clinical manifestations of Malaria



Clinical symptoms of malaria are caused by the lysis of red blood cells & the release of the parasite's merozoite form into the blood.

“Erythrocytic replication cycle”

Plasmodium lifecycle & antimalarial drugs



Classification of antimalarial drugs

classify based on:

- activities during the life cycle
- intention of use (prophylaxis & treatment)
- mechanism of actions

Chemoprophylaxis:

no antimalarial drug kills sporozoites



not truly possible to prevent infection

“ Drugs can only prevent the development of symptomatic malaria caused by the asexual erythrocytic forms ”

- **No single** antimalarial is effective against both liver & intra-erythrocytic stages of the life cycle that may co-exist in the same patient.
- **Complete elimination** of the parasite infection, requires **more than one drug**.

Susceptibility of antimalarial drugs at various developmental stages

GROUP	DRUGS	SPOROZOITE	LIVER STAGES		BLOOD STAGES	
			PRIMARY	HYPNOZOITE	ASEXUAL	GAMETOCYTE
1	Artemisinin	–	–	–	+	+
	Chloroquine	–	–	–	+	+/-
	Mefloquine	–	–	–	+	–
	Quinine, quinidine	–	–	–	+	+/-
	Pyrimethamine	–	–	–	+	–
	Sulfadoxine	–	–	–	+	–
	Tetracycline	–	–	–	+	–
2	Atovaquone/proguanil	–	+	–	+	+/-
3	Primaquine	–	+	+	–	+
	Tafenoquine	+	+	+	+	+

–, no activity; +/-, low to moderate activity; +, clinically important activity.

Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14th Edition.

Pharmacology of antimalarial drugs

(classified by mechanism of actions)

1. Inhibitors of heme metabolism:

chloroquine, quinine, quinidine,
mefloquine, artemisinin

2. Inhibitors of electron transport:

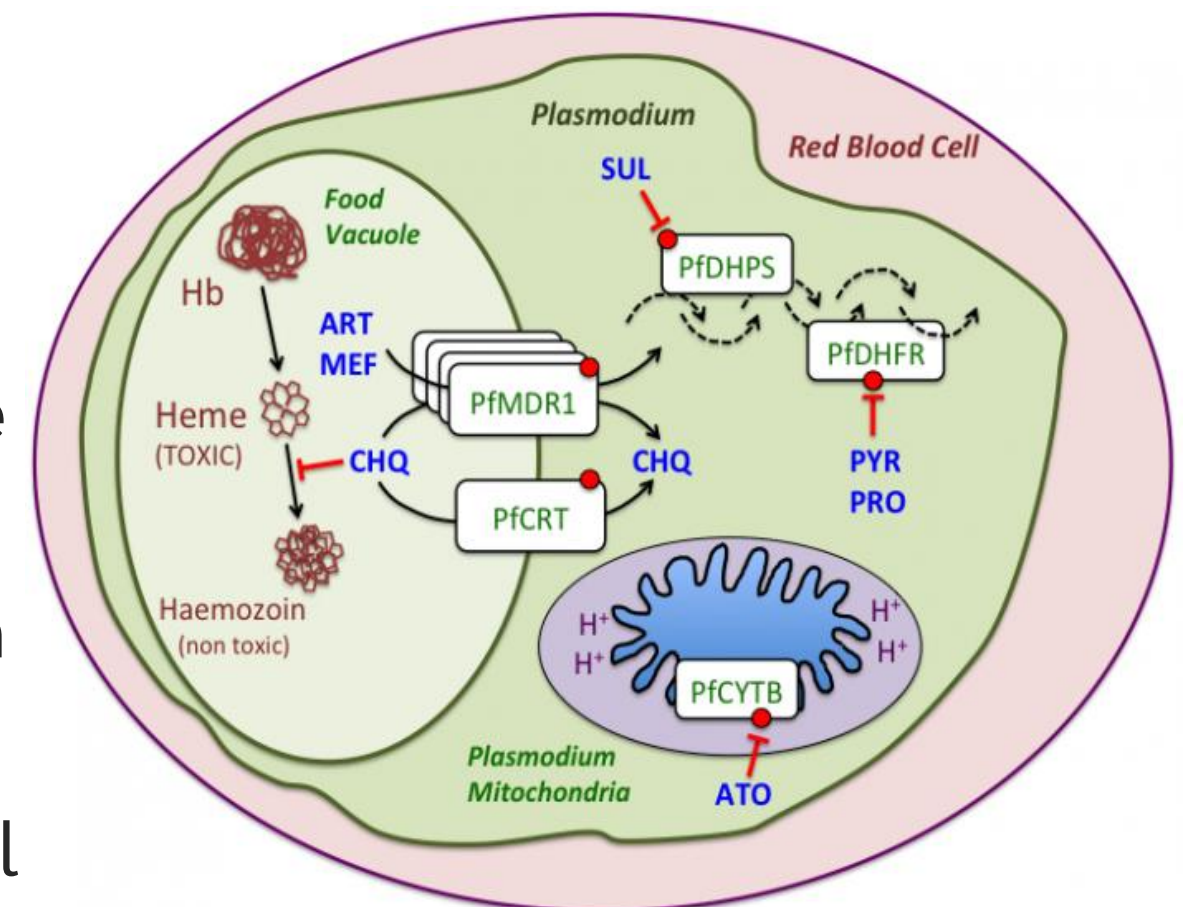
Primaquine, tafenoquine, atovaquone

3. Inhibitors of protein translation:

doxycycline, tetracycline, clindamycin

4. Inhibitors of folate metabolism:

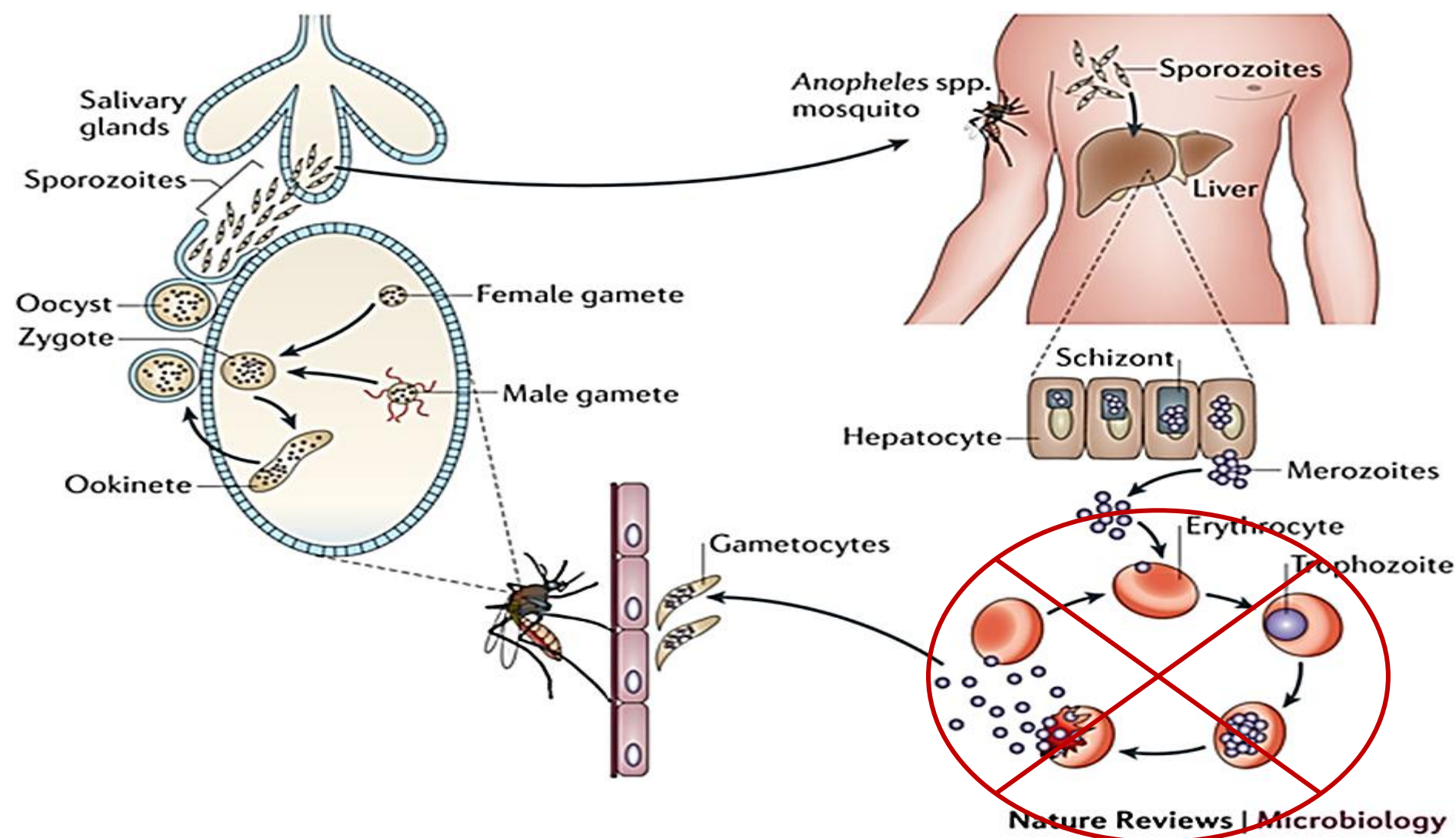
sulfadoxine-pyrimethamine, proguanil



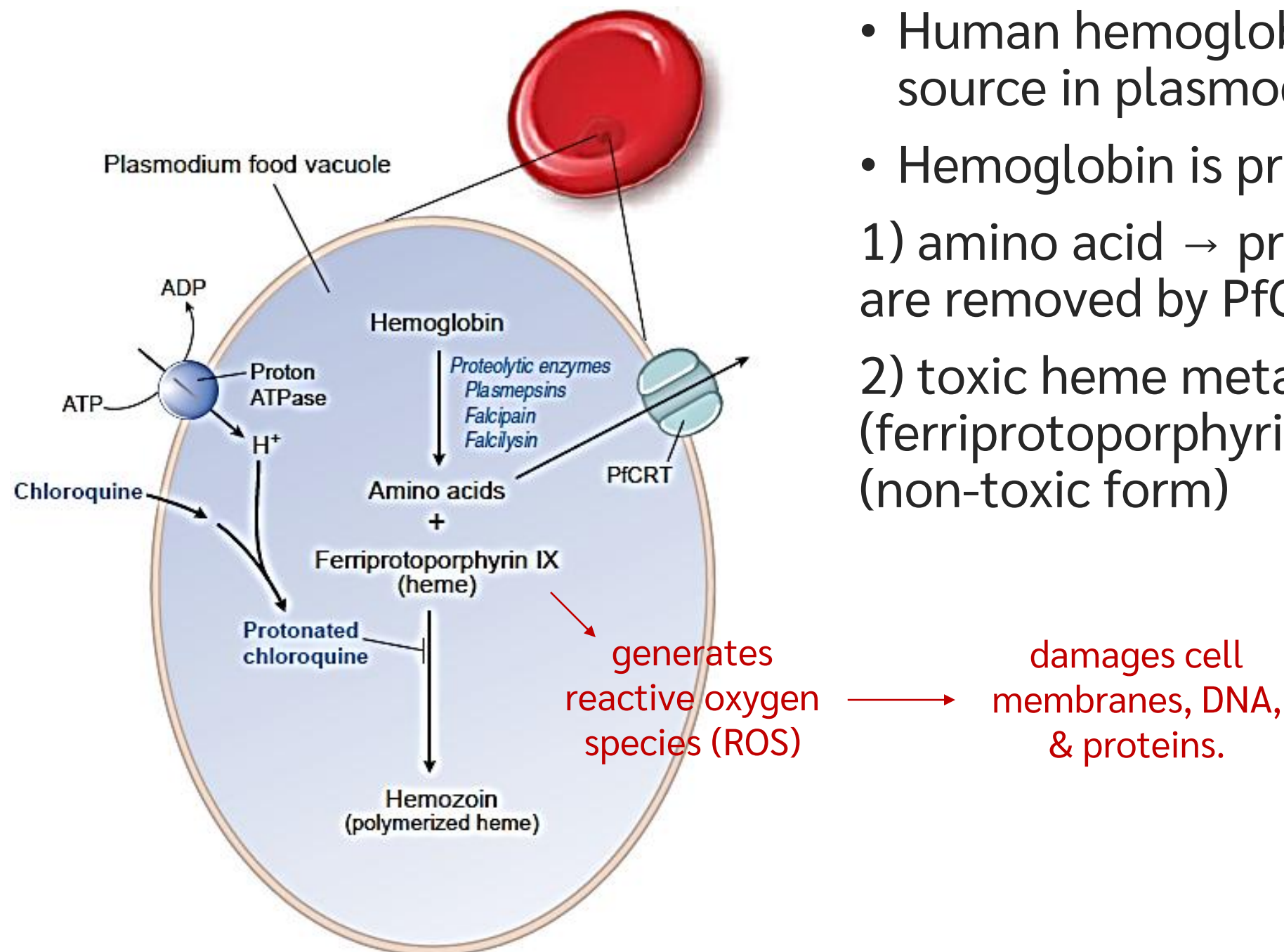
Chloroquine (CHQ), Artemisinin (ART),
Mefloquine (MEF), Sulfadoxine (SUL),
Pyrimethamine (PYR), Proguanil (PRO),
Atovaquone (ATO)

1. Inhibitors of heme metabolism

- Disrupt intra-erythrocytic malarial
- Chloroquine, Quinine and Quinidine, Mefloquine, Artemisinin



Mechanism of heme metabolism in the plasmodial food vacuole

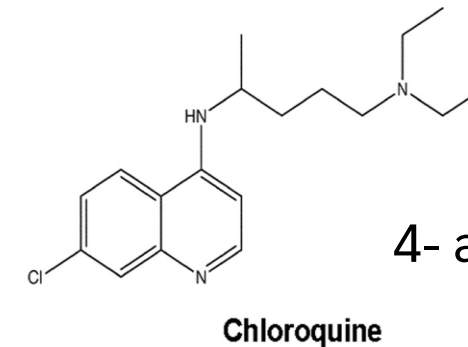


- Human hemoglobin is used as a food source in plasmodial vacuole.
- Hemoglobin is proteolyzed to
 - 1) amino acid → protonated amino acids are removed by PfCRT transporter.
 - 2) toxic heme metabolite (ferriprotoporphyrin IX) → hemozoin (non-toxic form)

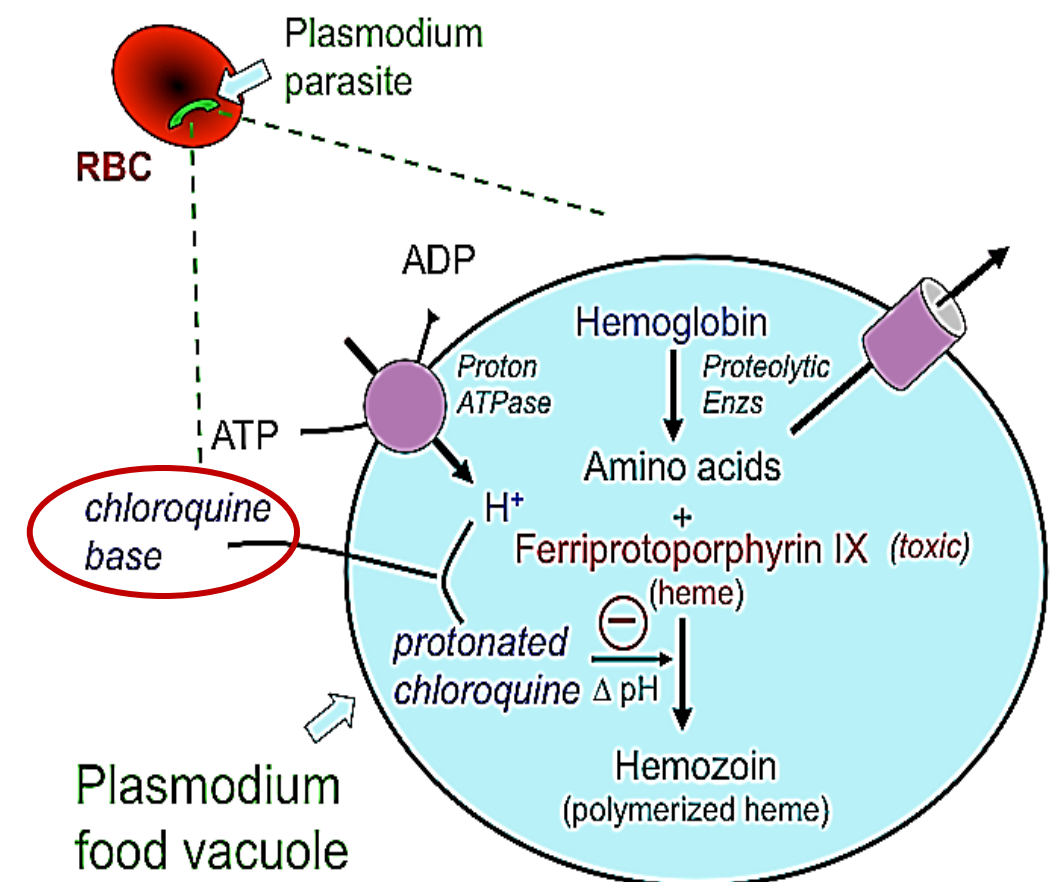
generates reactive oxygen species (ROS) → damages cell membranes, DNA, & proteins.

1.1 Chloroquine

- Derivatives: Piperaquine, Amodiaquine
- **MOA:** inhibit the detoxification of toxic products of hemoglobin
weak base → freely diffuses across the membrane of the parasite's food vacuole
↓
protonate inside vacuole - acidic environment (unable to diffuse out of the vacuole)
↓
bind to ferriprotoporphyrin IX & inhibit polymerization of heme metabolite
↓
accumulation of unpolymerized ferriprotoporphyrin IX
↓
oxidative stress, membrane damage, protein dysfunction

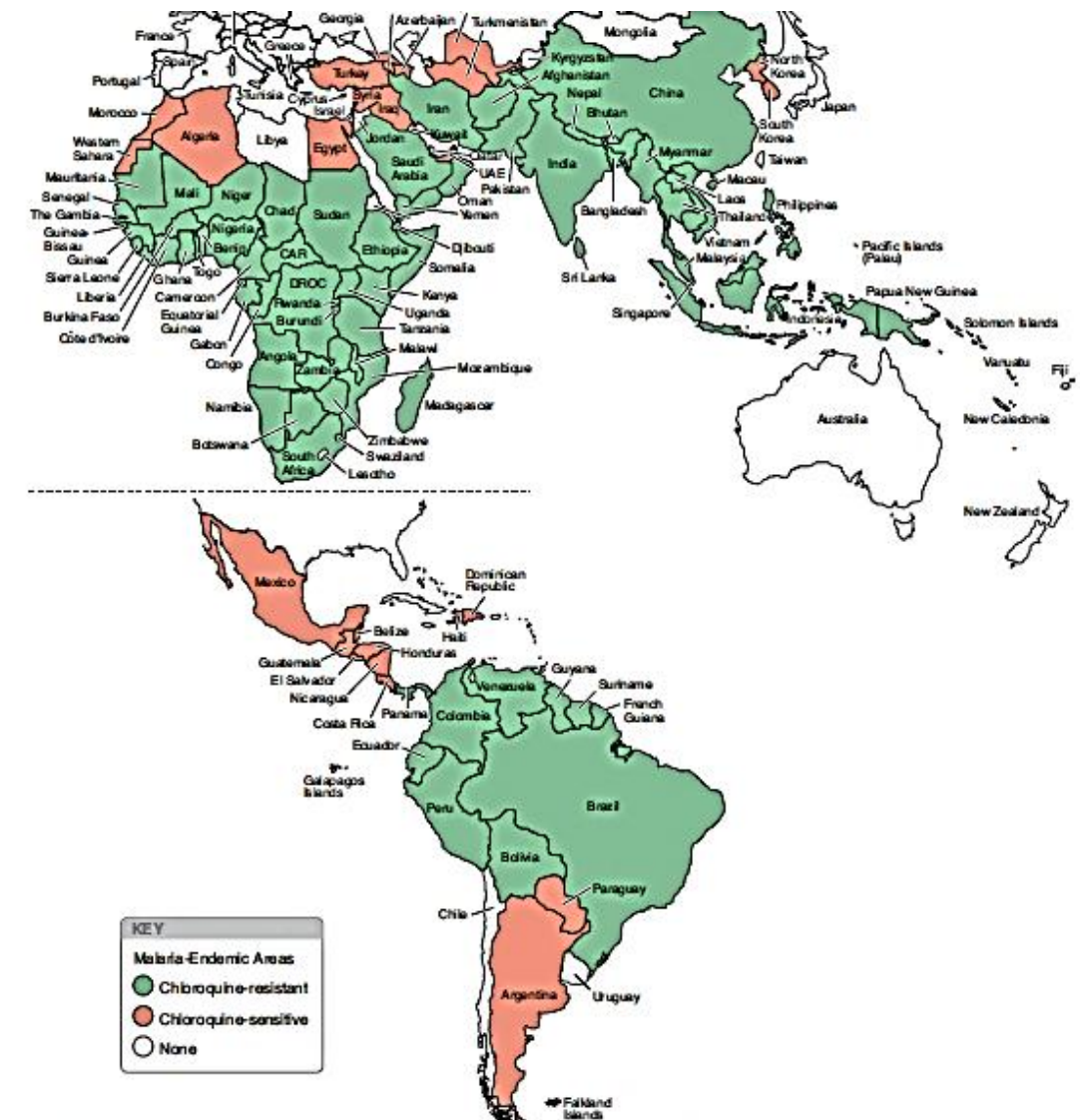


Structure:
4- aminoquinoline



Chloroquine resistance

- *P. falciparum* chloroquine resistance transporter encoded by *pfcr1* gene.
 - Located at food vacuole membrane
- P-glycoprotein transporter encoded by *pfmdr1*.
 - Located at parasite's plasma membrane
 - Contribute to chloroquine, quinine, artemisinin derivatives & mefloquine resistances





1.1 Chloroquine

Pharmacokinetics:

- Well absorbed from GI tract, peak plasma levels in ~3-5 hrs.
- Plasma protein binding ~ 60%
- Distribution: extensive tissue binding including CNS, placenta
- Metabolism: Hepatic CYP3A4/5, CYP2C8 -> desethylchloroquine (active) -> bisdesethylchloroquine
- Half-life: 30 - 60 days
- Renal excretion is increased by acidification of urine.



1.1 Chloroquine

Therapeutic uses:

- Highly effective against erythrocytic forms of
 - *P. vivax*,
 - *P. ovale*,
 - *P. malariae*,
 - *P. knowlesi*
 - *P. falciparum* (chloroquine-sensitive strain)
- Drug of choice for *P. ovale* & *P. malariae* chemoprophylaxis & treatment.
- *P. vivax*: resistant found in Asia & Pacific region but remains the drug of choice.
- *P. falciparum*: replaced by artemisinin-based combination therapies
- Non-malarial conditions:
 - Hepatic amebiasis,
 - rheumatoid arthritis,
 - systemic lupus erythematosus,
 - sarcoidosis



1.1 Chloroquine

Toxicity & ADRs:

- Safety margin is narrow (single dose of 30 mg/kg may be fatal).
- Acute chloroquine toxicity → high doses are administered too rapidly by parenteral routes.
- High doses → Toxic on CVD & CNS systems
 - CVD: hypotension, vasodilation, suppressed myocardial function, cardiac arrhythmias & eventual cardiac arrest
 - CNS: Confusion, convulsions & coma



1.1 Chloroquine

Toxicity & ADRs:

- High daily doses of chloroquine or hydroxychloroquine (>250 mg): irreversible retinopathy & ototoxicity.
 - Retinopathy: Drug accumulation in the retinal pigment epithelium (RPE) & bind to melanin within the retina -> cellular toxicity -> disruption of photoreceptor function -> retinal degeneration
 - Ototoxicity: Drug accumulation in the inner ear structures (e.g., cochlea) -> damage to auditory and vestibular hair cells
- Oral dose: GI upset, headache, visual disturbances & urticaria.
- Pruritus: most commonly among dark-skinned persons.



Aminoquinoline derivatives: Chloroquine, Amodiaquine, Piperaquine

Amodiaquine

- Closely related to chloroquine (mechanism and resistance)
- WHO lists artesunate plus amodiaquine as a recommended therapy for *P. falciparum* malaria.
- Toxicities: agranulocytosis, aplastic anemia, hepatotoxicity (do not use for long-term)

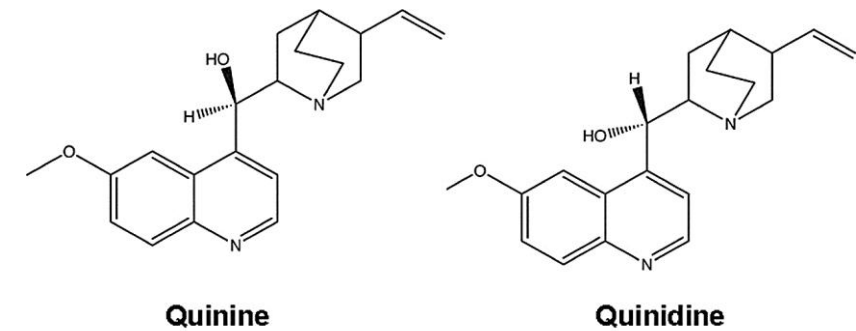
Piperaquine

- Piperaquine combined with dihydroartemisinin for the treatment of *P. falciparum* malaria
- Long half-life (28 d) than amodiaquine (14 d), mefloquine (14 d), lumefantrine (4 d)
- Advantageous in high-transmission areas.
- Dihydroartemisinin-piperaquine: 1st line therapy for uncomplicated *P. falciparum* malaria in some countries in Asia (Thailand).

Amodiaquine & Piperaquine are key components of artemisinin-based combination therapies (ACTs)

1.2 Quinine and Quinidine

- Alkaloid of cinchona (bark)
- Acts against asexual erythrocytic forms.
- No significant effect on hepatic forms of malarial parasites.



MOA:

- Bind with heme and prevent detoxification.
- Intercalate into DNA -> inhibiting DNA strand separation, transcription & translation -> decrease growth & replication of the erythrocytic form of plasmodia.
- Resistance mechanism: *pfCRT* point mutations



1.2 Quinine and Quinidine

Pharmacokinetics:

- readily absorbed when given orally or intramuscularly
- T_{max} : 3-8 hrs., V_d : 1.5 L/kg, $T_{1/2}$: 11 hrs.
- Metabolized extensively by hepatic CYP3A4
 - 3-hydroxyquinine (major metabolite of quinine) possibly causes toxicity
- Excreted in an unaltered form in urine
- Acidic urine can increase renal excretion.

Therapeutic uses:

- Acute blood-stage malaria
- But not used for prophylactic (toxicity & short $T_{1/2}$)
 - more toxic & less effective than chloroquine
- Non-antimalarial: treatment of nocturnal leg cramps



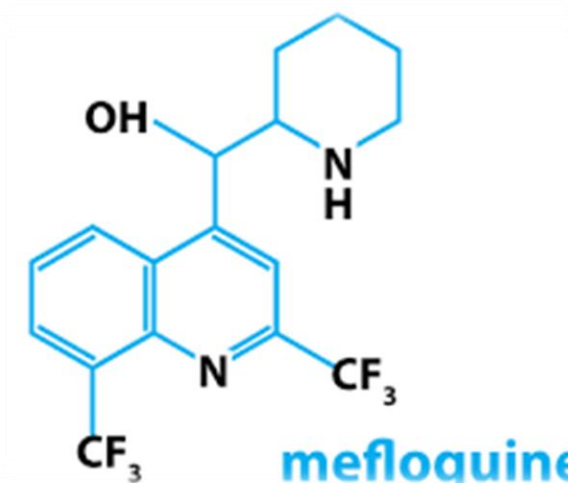
1.2 Quinine and Quinidine

ADRs:

- Triad of dose-related (full therapeutic/excessive doses)
 - cinchonism
 - hypoglycemia
 - hypotension
- Cinchonism: tinnitus, high-tone deafness, headache, nausea, vomiting, visual disturbances, postural hypotension
- GI symptoms: nausea, vomiting, abdominal pain & diarrhea
- Prolong cardiac QT interval (rare)

1.3 Mefloquine

- Unknown exact mechanism: may disrupt polymerization of hemozoin
- Highly effective blood schizonticide.
- Reserved for the prevention & treatment of malaria caused by drug resistant *P. falciparum* & *P. vivax*
- Long half-life (20 d) & highly protein-bound, extensively distributed in tissues and eliminated slowly -> single-dose treatment regimen
- More effective in combination with artemisinin compounds.
- Should not be co-administered with quinine, quinidine, or halofantrine.





1.3 Mefloquine

- Metabolism: mainly liver by CYP3A4
- Excretion: mainly fecal route & 10% of unchanged form excreted via urine

ADRs: (consider balance of risk vs benefit)

- Nausea, vomiting, dizziness
- Cardiac conduction abnormalities: bradycardia, prolongation of the QT interval & arrhythmia
- FDA black box warning (2013):
 - potential neurologic & psychiatric toxicities (vivid dreams/nightmares (common), insomnia, anxiety, depression, hallucinations, seizures, psychosis)

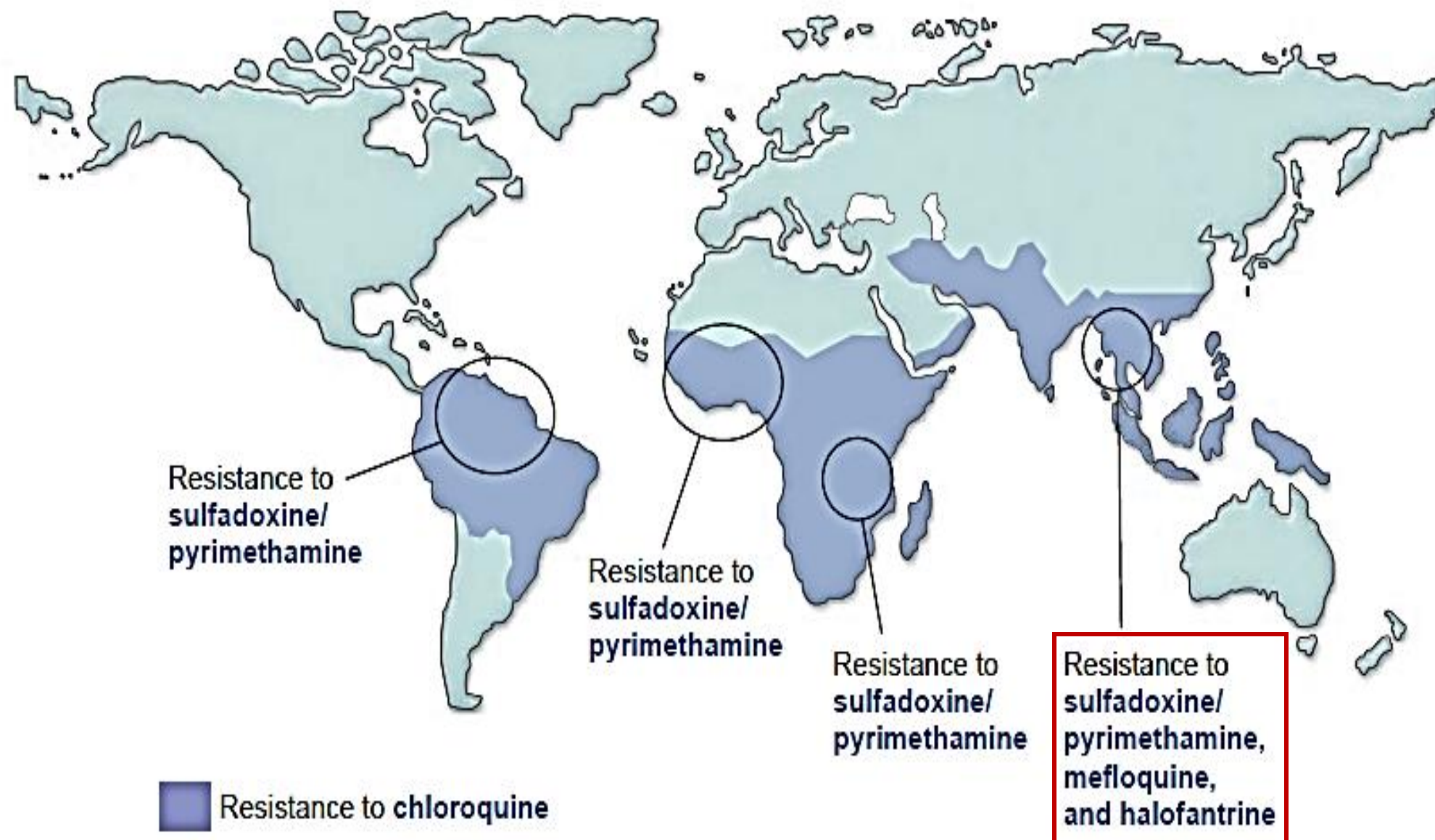
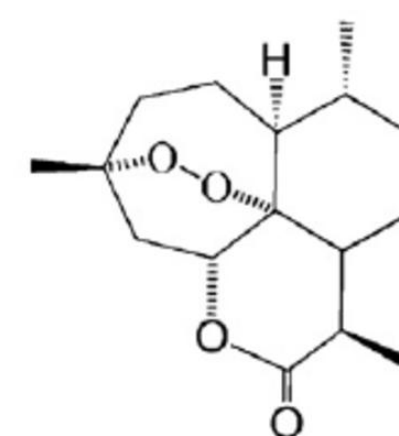


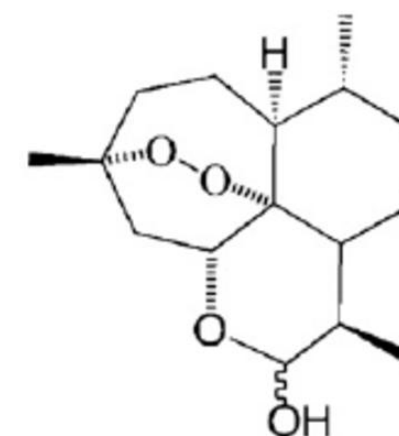
FIGURE 36-4. Geographic distribution of drug-resistant *Plasmodium falciparum*. Historically, chloroquine has been the drug of choice for prophylaxis and treatment of individuals with *P. falciparum* malaria. Unfortunately, *P. falciparum* is now resistant to chloroquine in most areas of the world (*blue shading*). In many areas, *P. falciparum* is also resistant to other antimalarial agents, including sulfadoxine–pyrimethamine, mefloquine, and halofantrine. (Halofantrine is associated with potentially lethal cardiac toxicity and is therefore seldom used.)

1.4 Artemisinin & derivatives

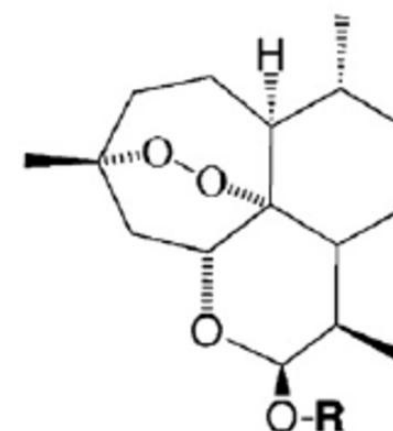
- Derived from *Qing hao* (*Artemisia annua*)
- Structures: both sesquiterpene lactone and a cyclic endoperoxide
- Very potent and fast-acting antimalarials (rapid parasite clearance and fever resolution)
- Affect blood-stage gametocytes → decrease transmission of malaria from an infected human.
- Not effective as a prophylactic agent against malaria.
- Do not use as monotherapy.



Artemisinin



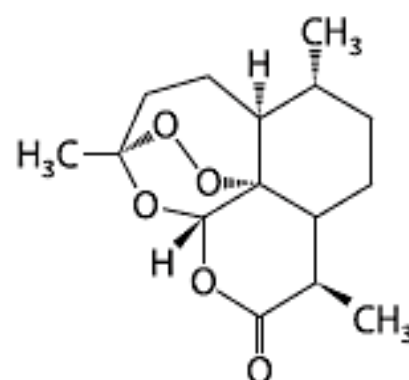
Dihydroartemisinin



R: -CH₃ (**Artemether**)
R: -CH₂-CH₃ (**Arteether**)
R: -CO-CH₂-CH₂-COOH
(**Artesunate**)

True resistance to artemisinin has not been reported but reflects delayed parasite clearance time.

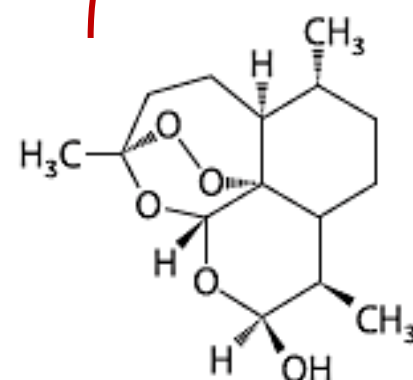
Analogs have been synthesized to increase solubility and improve antimalarial efficacy.



ARTEMISININ

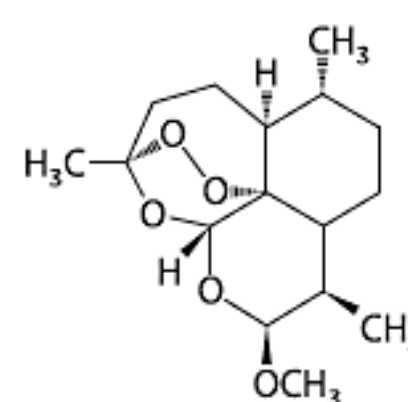


- insoluble & can only be used orally but low BA
- Formulations: oral & rectal administration
- T_{1/2}: 3-5 hrs.



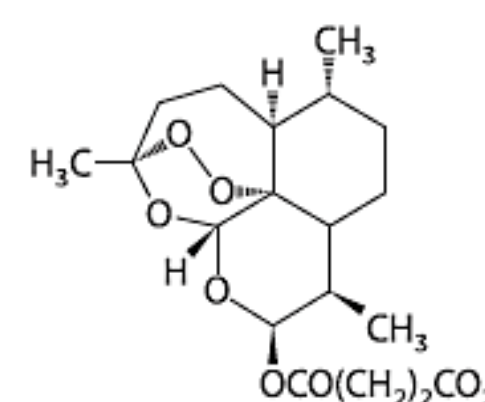
DIHYDROARTEMISININ

- water-soluble; oral administration
- T_{1/2}: 30-60 min



ARTEMETHER

- lipid-soluble; oral, IM & rectal administration
- T_{1/2}: 2-3 hrs.

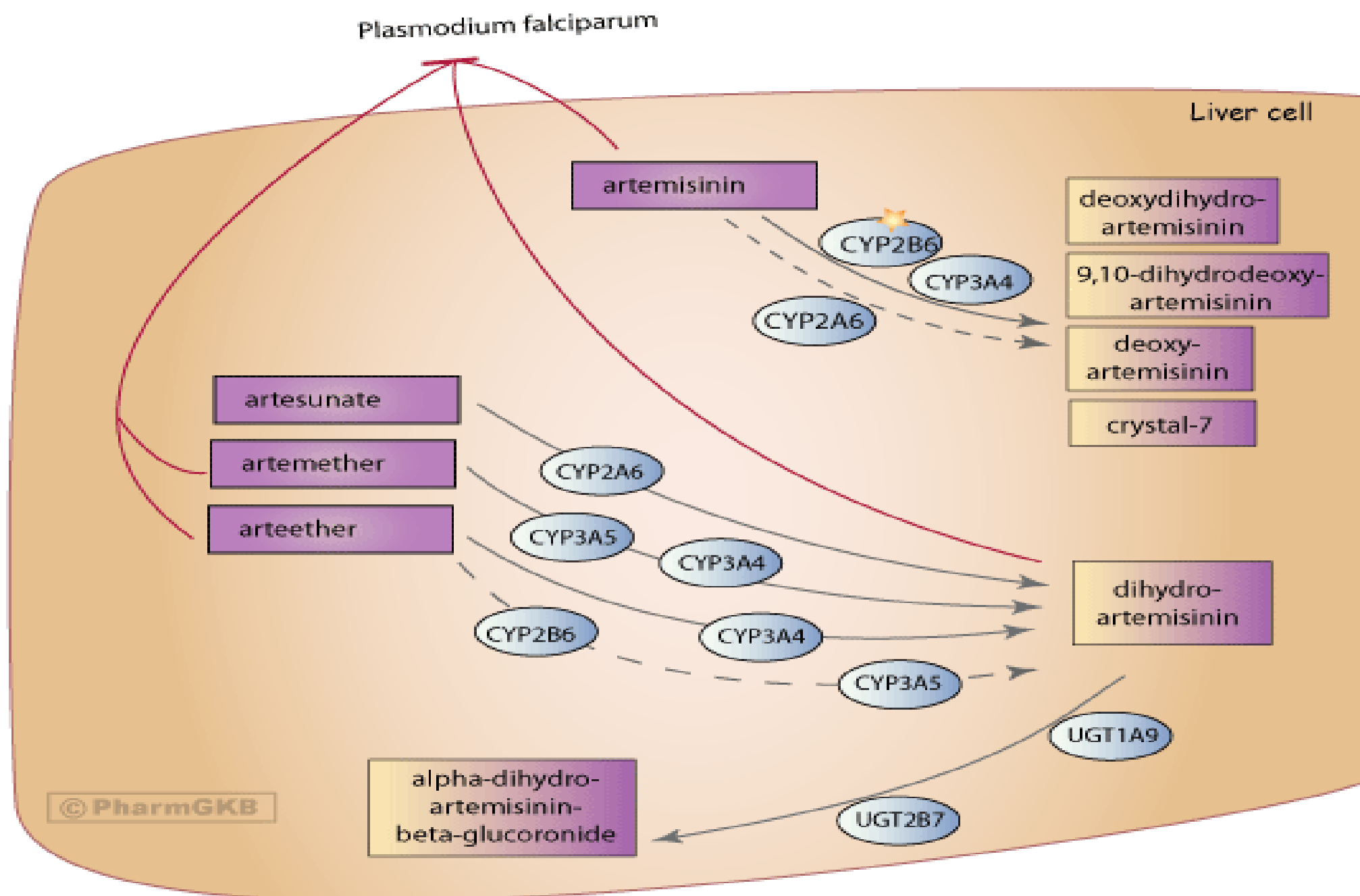


ARTESUNATE

- water-soluble; oral, IV, IM, & rectal administrations
- T_{1/2}: 30-60 mins

Artemisinin, artesunate, artemether $\xrightarrow{\text{metabolized}}$ **dihydroartemisinin**

Artemisinin & artesunate are CYPs 2B6 and 3A4 inducers



Mechanism of Artemisinin

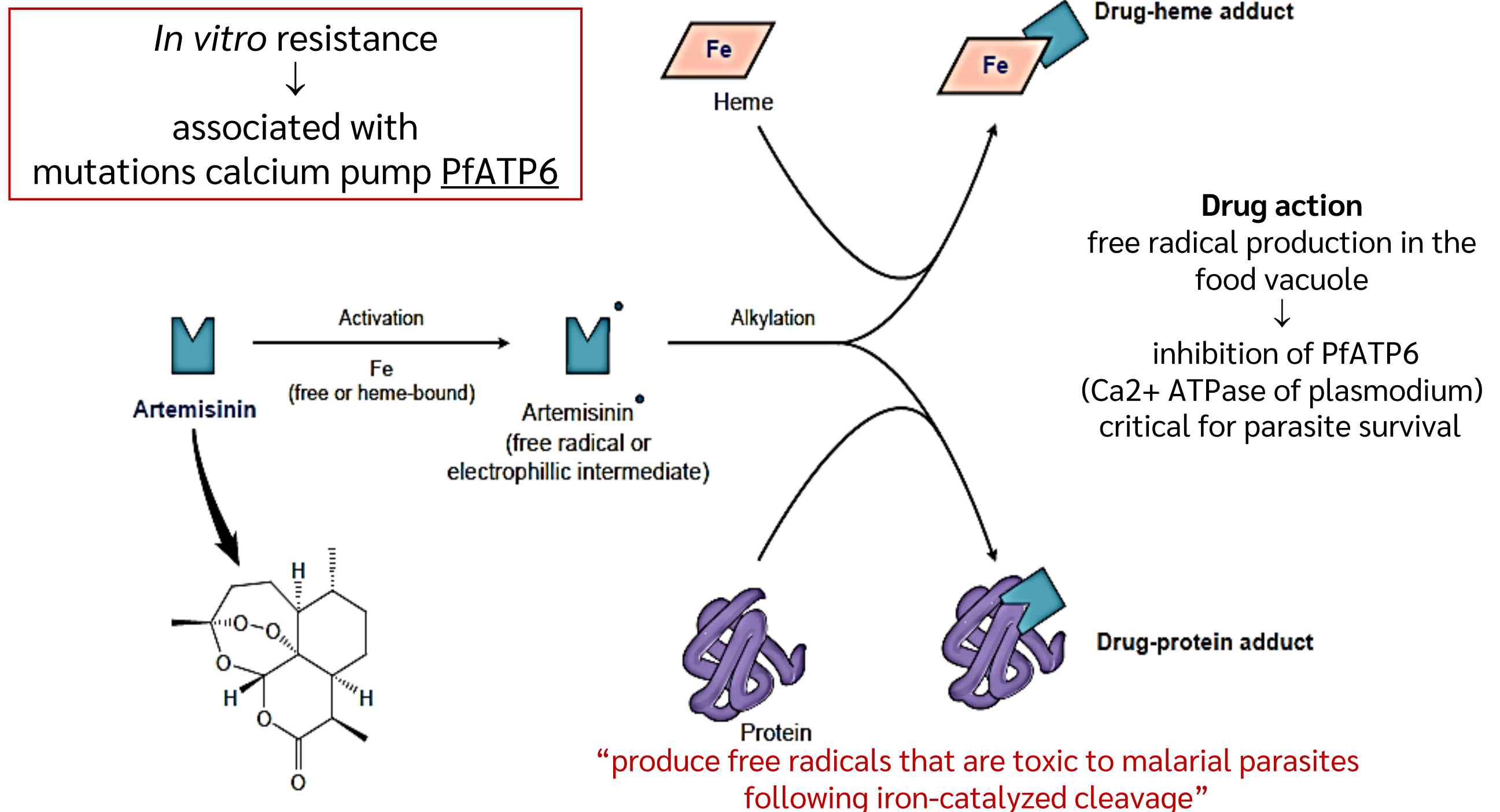


FIGURE 36-5. Proposed mechanism of action of artemisinin. Artemisinin is a cyclic endoperoxide that forms a free radical after activation by Iron (Fe). The mechanism of action of artemisinin is not known with certainty, but may involve alkylation of macromolecules such as heme and proteins, resulting in the formation of artemisinin–heme adducts and artemisinin–protein adducts that are toxic to plasmodia. One such adduct may involve PfATP6, a parasite Ca²⁺ ATPase (*not shown*).



1.4 Artemisinin & derivatives

Therapeutic uses:

- WHO strongly recommends not to use as monotherapy.
- FDA: Artemether-lumefantrine for oral treatment of uncomplicated *P. falciparum* (Coartem regimen)
- **Artemisinin-based Combination Therapies (ACTs)**
 - Fixed combinations (usually with a second agent with a longer half-life)
 - Increase efficacy (due to short $T_{1/2}$) & reduce drug resistance.
 - Ex: artemether–lumefantrine, artesunate–amodiaquine, artesunate–pyronaridine,, dihydroartemisinin–piperaquine



1.4 Artemisinin & derivatives

Toxicity & ADRs:

- Better tolerated than other antimalarial agents.
- Artemisinin: potential toxic effects in infants & during the first trimester of pregnancy.
 - Chloroquine, quinine + clindamycin is recommended during the first trimester.
 - Not to use ACTs for the treatment of children ≤ 5 kg or during the first trimester of pregnancy.
- Other toxicities (rare):
 - Intramuscular injection of oil-based formulations of Artemisinin: brainstem neuropathy (animal)
 - Auditory impairment & other neurotoxic effects in humans.
 - Allergic reaction, neutropenia, anemia, hemolysis, elevated liver enzymes.



ACTs Partner Drugs

Drug	Indication	Pharmacokinetics	Note
Lumefantrine	uncomplicated malaria	• $T_{1/2}$: 4-5 d, • large V_d, • high-fat meals increase absorption	Coartem [®] (artemeter + lumefantrine)
Amodiaquine	uncomplicated <i>P. falciparum</i>	• Rapidly converted by hepatic CYPs to active metabolite • $T_{1/2}$: 9-18 d	Artesunate + Amodiaquine
Piperaquine	chloroquine resistance <i>P. falciparum</i>	• Large V_d • Rates of excretion reduce after multiple doses • Longest plasma $T_{1/2}$: 5 weeks	Dihydroartemisinin + piperaquine



Plasma half-lives of drugs used in artemisinin-based combination therapies

Antimalarial	$t_{1/2}$ of artemisinin derivative	$t_{1/2}$ of partner drug	Regions currently in use*
Artemether–lumefantrine	~3 hr	4–5 days	Africa, EM, SE Asia, WP and SA
Artesunate–mefloquine	<1 hr	14–21 days	Africa, SE Asia, WP and SA
Artesunate–amodiaquine	<1 hr	9–18 days [‡]	Africa and EM
Dihydroartemisinin–piperaquine	45 min	~5 weeks	SE Asia
Artesunate–pyronaridine [§]	<1 hr	16 days	NA
Chloroquine	NA	1–2 months	Africa, EM, SE Asia, WP and SA
Sulphadoxine–pyrimethamine	NA	~4 days (S) or ~8 days (P)	Africa, EM (IPT in Africa, EM and WP)

* Data from REFS 106,140.

[‡] This refers to the $t_{1/2}$ of the active metabolite monodesethylamodiaquine; the $t_{1/2}$ of amodiaquine is ~3hr.

[§] Recently completed Phase III trials.

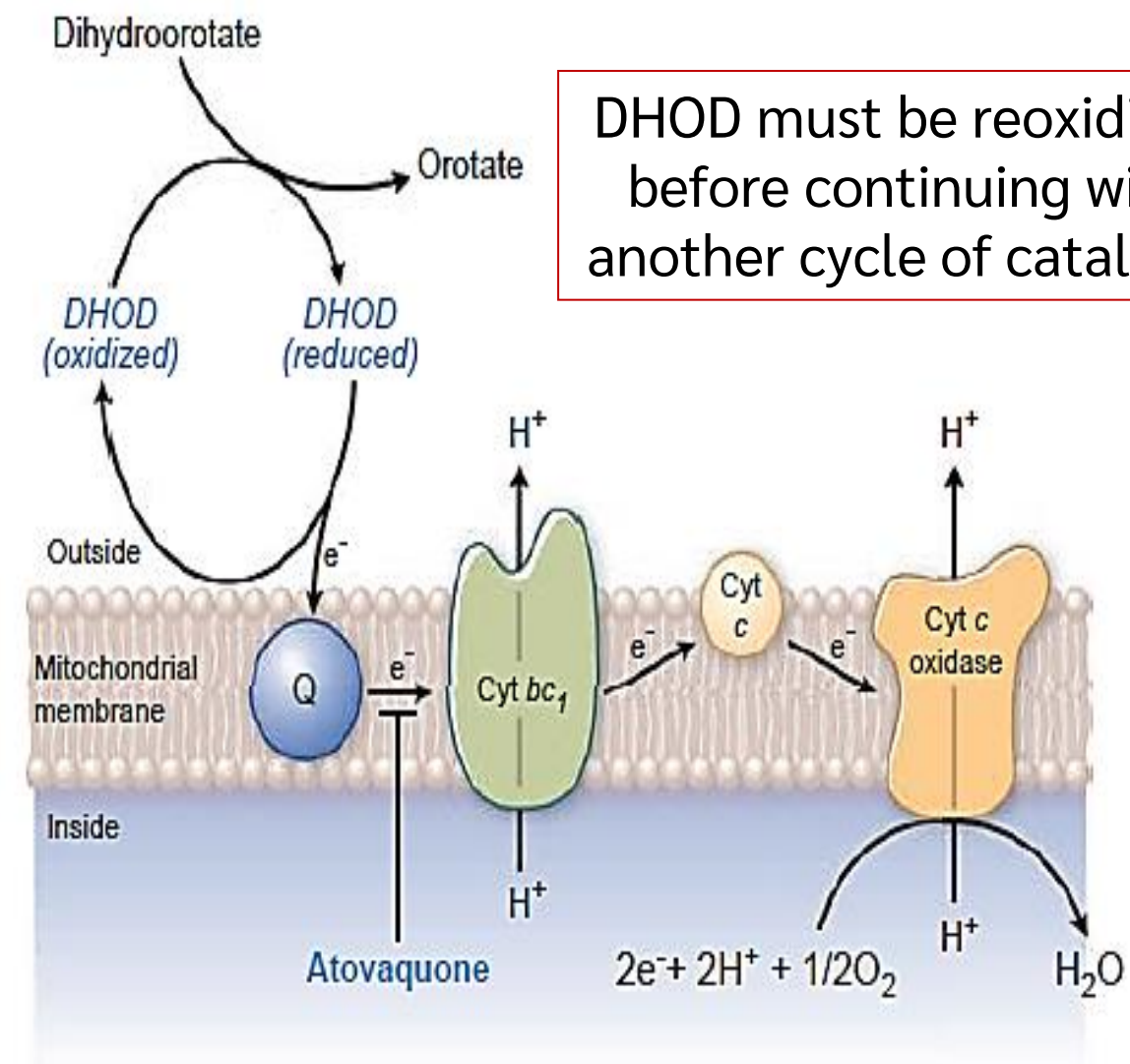
^{||} These former first-line antimalarials are included as a reference. EM, eastern Mediterranean; IPT, intermittent preventive treatment; NA, not applicable; P, pyrimethamine; S, sulphadoxine; SA, South America; SE Asia, Southeast Asia; $t_{1/2}$, half-life; WP, Western Pacific.

2. Inhibitors of Electron Transport

- “Electron transport chain”
- Plasmodia uses ATP (source of energy) directly from glycolysis not using mitochondrial electron transport

But

- Use electron transport for oxidation of key enzymes involved in nucleotide synthesis.
- Dihydroorotate dehydrogenase (DHOD): pyrimidine synthesis for DNA replication



DHOD must be reoxidized before continuing with another cycle of catalysis.

2.1 Primaquine

- Drug of choice for eradicating dormant liver forms of *P. vivax* & *P. ovale*

Activity:

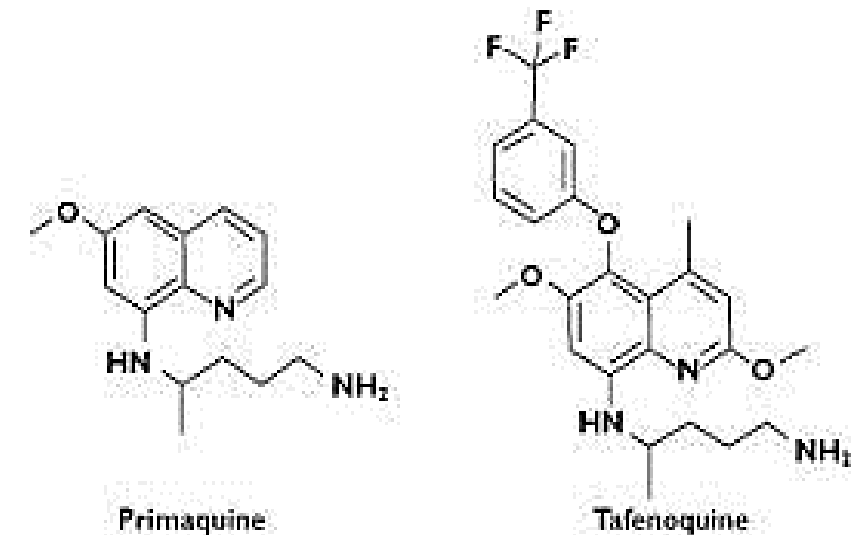
1. Exoerythrocytic tissue stages

- primary and latent hepatic stages of *Plasmodium spp.* and prevents relapses in *P. vivax* & *P. ovale* infections

2. Gametocytocidal activity: against *P. falciparum* and other Plasmodium species, inactive against asexual blood stage parasites.

MOA:

- Interfering with mitochondrial electron transport in the parasite (interfering with the function of ubiquinone)
- Converted to electrophilic intermediates -> generating reactive oxygen species



2.1 Primaquine

Pharmacokinetics

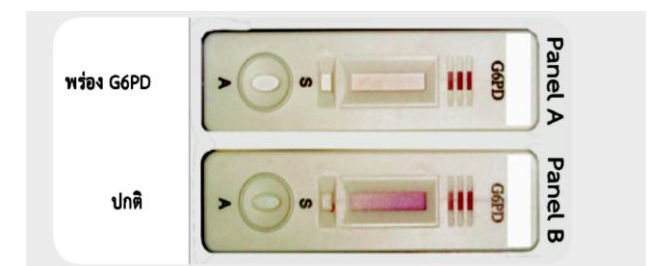
- Complete absorption (100%)
- T_{max} : 3 hrs
- $T_{1/2}$: 7 hrs
- Extensive tissue distribution
- CYP1A2 inducer (caution to use with warfarin)

Contraindications:

- G6PD deficiency → hemolysis
- G6PD should be screened before primaquine use.
- Pregnancy (fetus is relatively G6PD-deficiency)
- Patient with a history of methemoglobinemia.

Toxicity and ADRs

- Mild to moderate abdominal distress (prevented by taking with meal).
- Mild anemia, hemolysis, “methemoglobinemia” & leukocytosis (high dose)



New drug: Tafenoquine

- Long-acting analog of primaquine (approved by USFDA in 2018)
- T_{1/2}: 15 d
- Activity against pre-erythrocytic (liver) & erythrocytic (asexual) forms, gametocytes of *P. vivax* and *P. falciparum*.
 - radical cure of *P. vivax* malaria (aged ≥ 16 years)
 - prophylaxis of malaria in patients aged ≥ 18 years

MOA:

- interfere with mitochondrial function leading to the apoptotic-like death of the organism (like primaquine)
- inhibit haematin polymerization (like chloroquine)

การจ่ายยา Tafenoquine

สามารถจ่ายยาในผู้ป่วยโรคไข้มาลาเรียจากเชื้อ *P. vivax* และ *P. ovale* ที่อายุ 16 ปีขึ้นไปเท่านั้น ในขนาด 300 มก. รับประทานครั้งเดียว โดยจ่ายร่วมกับ Chloroquine ดังนี้

วันนับจากวันที่เริ่มรักษา	Day 0 (เริ่ม)	Day 1	Day 2
Chloroquine	1 ครั้ง (600 มก.)	1 ครั้ง (600 มก.)	1 ครั้ง (300 มก.)
Tafenoquine*	1 ครั้ง (300 มก.)	-	-

หมายเหตุ ห้ามยา Tafenoquine ในหญิงตั้งครรภ์โดยเด็ดขาด



New drug: Tafenoquine

Therapeutic uses:

1. Therapy (radical cure) of acute *P. vivax* and *P. ovale* malaria
 - May be given together with a blood schizontocide (usually chloroquine) to eradicate erythrocytic stages and reduce drug resistance
2. Chemoprophylaxis of malaria
 - CDC: daily treatment with 30 mg (0.5 mg/kg) of primaquine base provided good protection against *P. falciparum* and *P. vivax* malaria
3. Gametocidal action



New drug: Tafenoquine

ADRs:

- Common: headache, dizziness, back pain, diarrhea, nausea, motion sickness, depression, anxiety, abnormal dreams & an elevated alanine aminotransferase level
- Hemolytic anemia in G6PD (like primaquine)

Precautions and contraindications

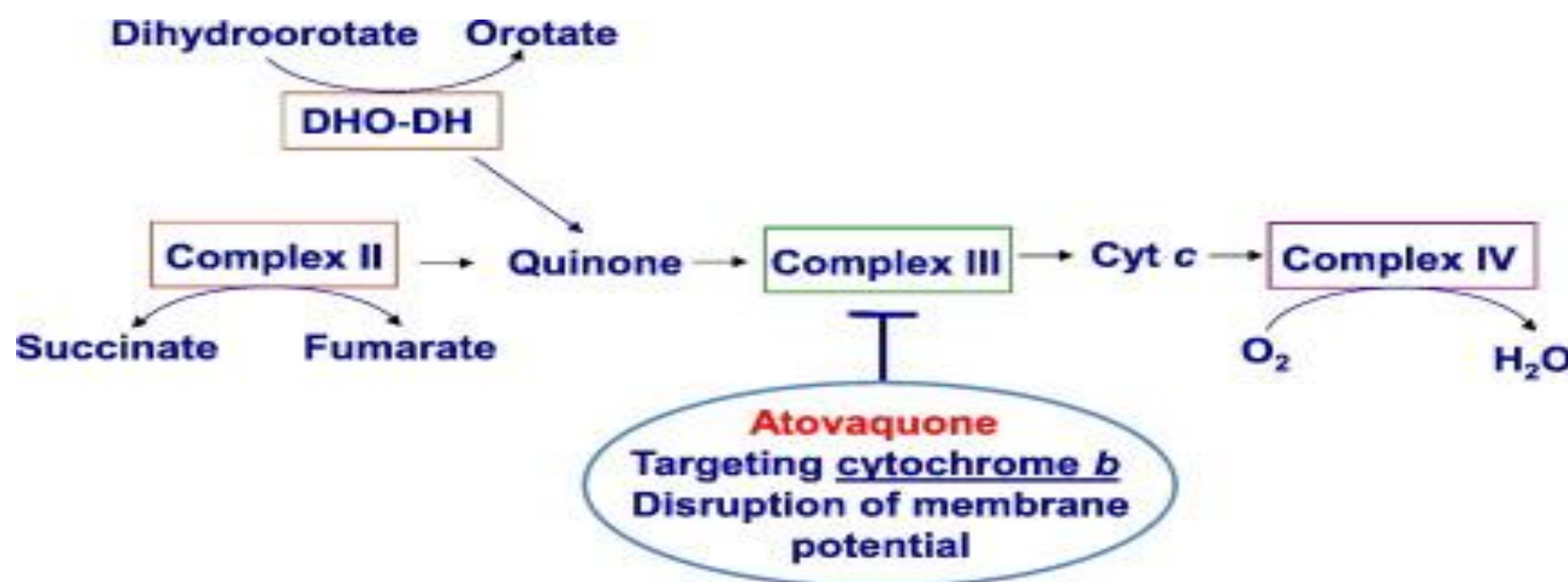
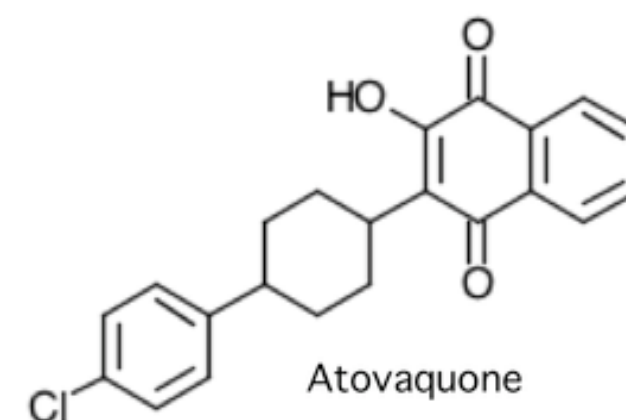
- G6PD deficiency or unknown G6PD status
- Hypersensitivity to primaquine (chemical similarity to primaquine)

2.2 Atovaquone

- Effective against liver stages of *P. falciparum* but not *P. vivax* liver stage hypnozoites.

MOA:

- Selectively inhibit the interaction between reduced ubiquinone and cytochrome bc1 complex → collapse the mitochondrial electron transport → pyrimidine synthesis



2.2 Atovaquone

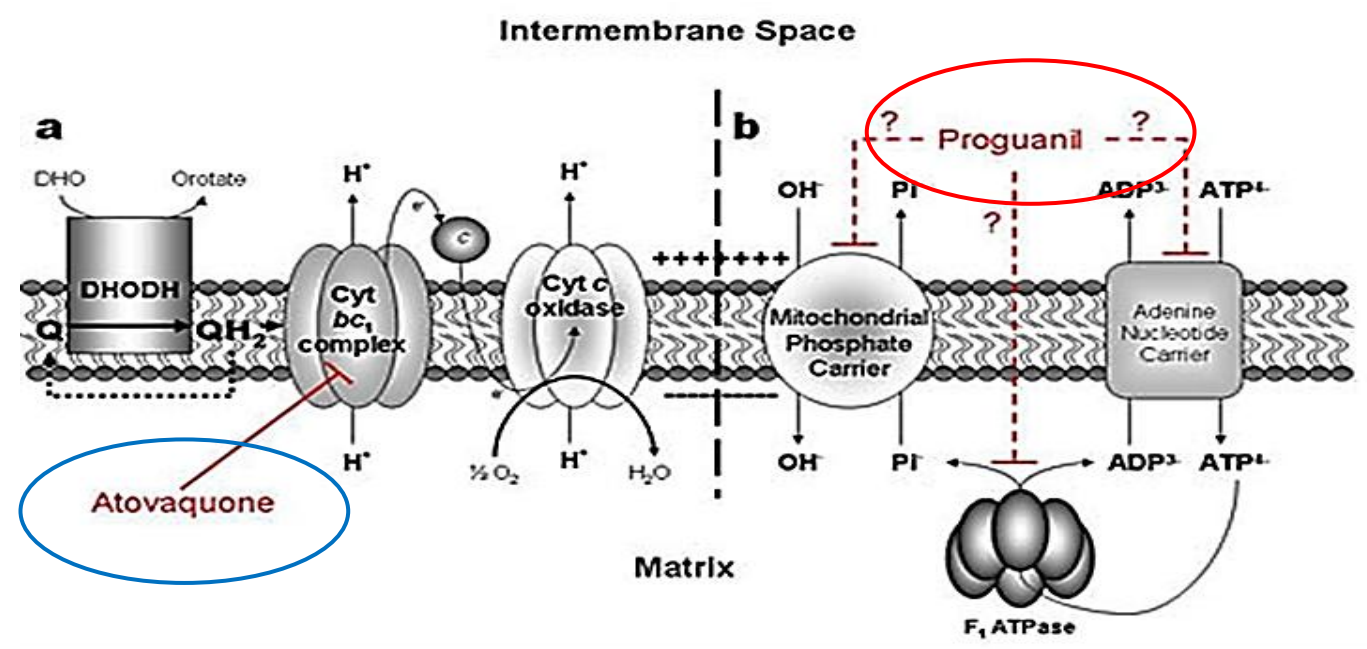
Mechanism of resistance:

Mutation in the cytochrome *bc1* complex

-> “Do not use as single agent”

Often co-administer with doxycycline (protein synthesis inhibitor) or proguanil (dihydrofolate reductase inhibitor)

-> “Malarone: atovaquone + proguanil”





2.2 Atovaquone

Therapeutic uses:

- Atovaquone-proguanil
(fixed dose 250 mg: 100 mg)
- Chemoprophylaxis in adults
and children ≥ 11 kg
- Treatment of uncomplicated
P. falciparum malaria in adults
and children ≥ 5 kg

Precautions & contraindications:

- Children weighing < 11 kg
- pregnant women & lactating mothers.
- Concomitant with rifampin and tetracycline.

ADRs:

- Abdominal pain
- Nausea, vomiting, diarrhea
- Headache
- Rash
- Transient elevations of serum
transaminase or amylase (rare)



3. Inhibitors of Translation

Doxycycline/ tetracycline

- MOA: Disrupt protein synthesis of parasite by binding to 30s ribosome subunit.
- Use in combination with artesunate & quinine in chloroquine-resistant *P. falciparum*
- Use as a standard chemoprophylaxis, especially in Southeast Asia with high rates of resistance to mefloquine.
- ADRs : photosensitivity, tooth discoloration, vaginal candidiasis, GI side effects

Clindamycin

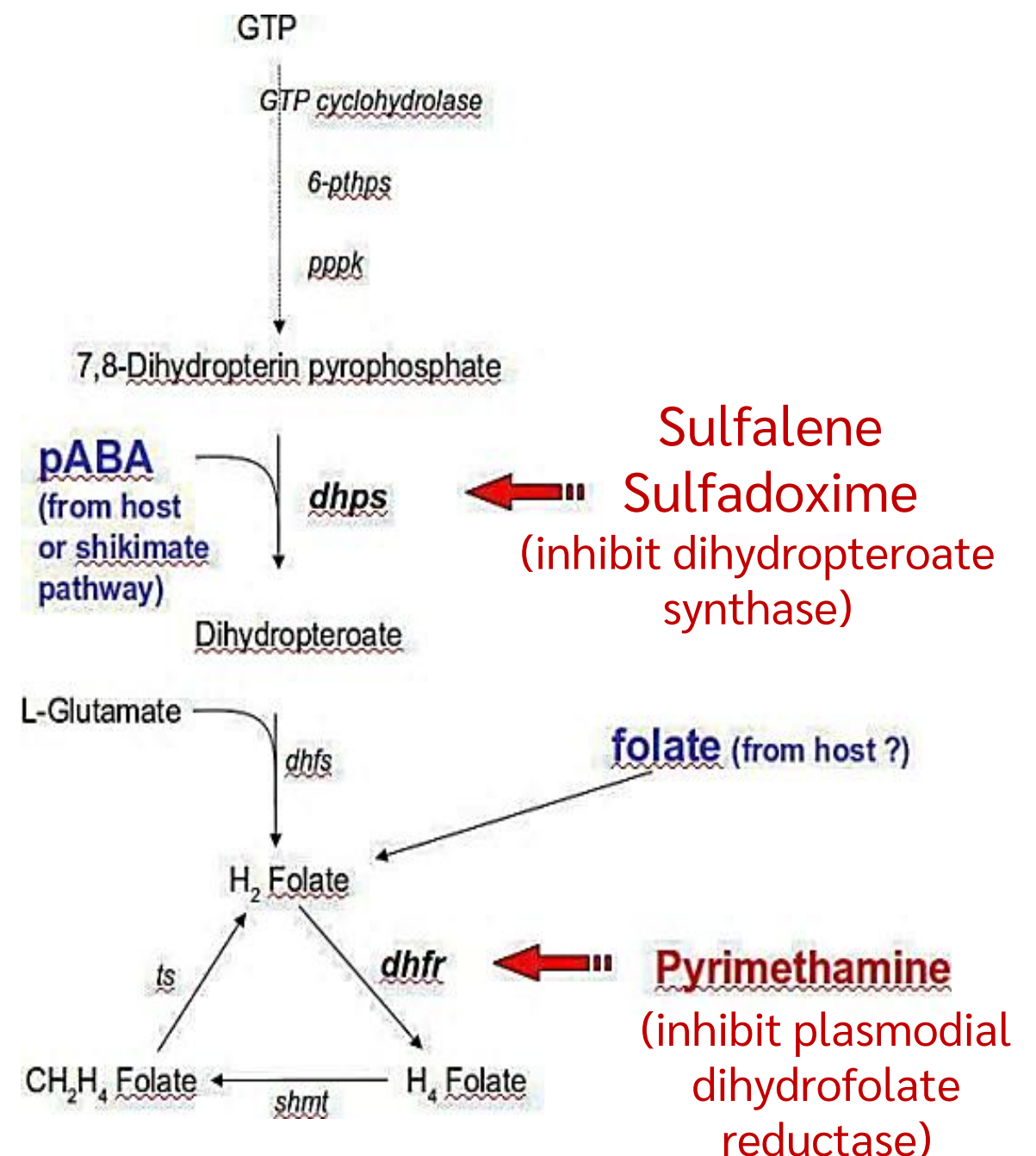
- MOA: inhibit protein synthesis of the parasite by binding to 50s ribosome subunit.
- Use instead of tetracycline & doxycycline (in pregnant woman & children < 8 years old) in combination with artesunate and quinine in chloroquine-resistant *P. falciparum*
- ADRs: increased risk of antibiotic-associated diarrhea and colitis caused by *Clostridium difficile*

4. Inhibitors of Folate Metabolism

4.1 Sulfalene–pyrimethamine

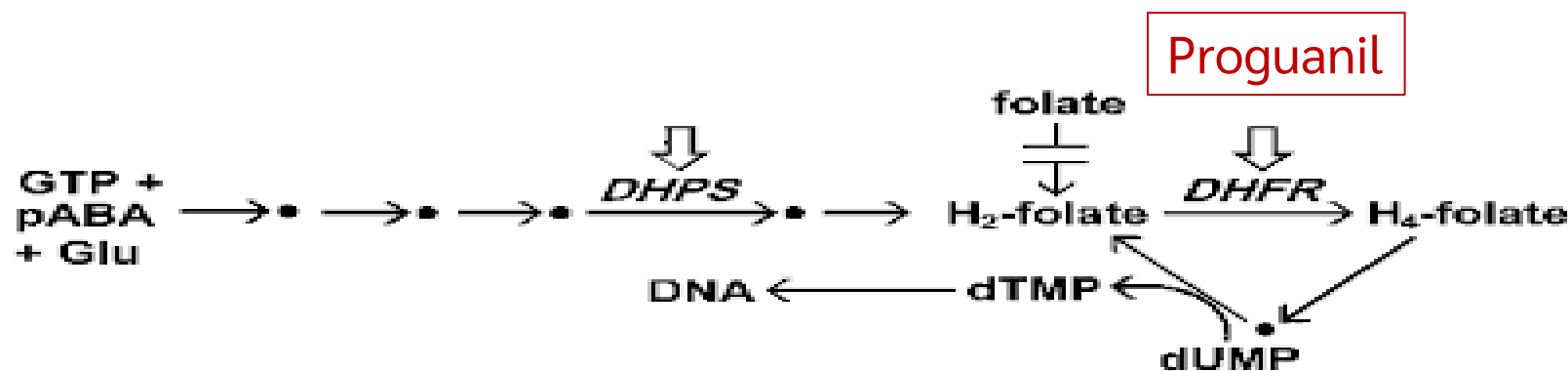
- Inhibit growth of the malarial parasite
- Effective against blood schizont stages of *P. falciparum* (not against gametocytes).
- Highly protein-bound $\rightarrow$ long $T_{1/2}$
- Recently found increasing resistance Fansidar regimen: fixed combination of sulfadoxine (500 mg) + pyrimethamine (25 mg)
- Slower-acting than other available agents $\rightarrow$ no longer a recommended therapy for malaria.

Folate pathway in Plasmodium parasites



4.2 Proguanil

- **MOA:** selective inhibitor of the plasmodial dihydrofolate reductase → inhibit purine and pyrimidine synthesis



- Effective against the hepatic, pre-erythrocytic forms of *P. falciparum* & *P. vivax*
- Used in a synergistic combination with atovaquone (once daily for 3 days, to treat drug-resistant strains of *P. falciparum* or *P. vivax*)
- **ADRs:** oral ulcerations, pancytopenia, thrombocytopenia, granulocytopenia
- Safe for use during pregnancy (safe when used in conjunction with chloroquine, atovaquone, tetracyclines and other antifolates)



Other drugs

Halofantrine

- inhibit polymerization of heme.
- effective against erythrocytic.
- rapidly effective against *P. falciparum*
- Cardiac toxicity -> limited usage
- Embryotoxic in animals -> contraindicated in pregnancy

Lumefantrine

- inhibit polymerization of heme.
- available only as a fixed-dose combination with artemether (Coartem®)
- 1st line therapy for uncomplicated *P. falciparum* malaria
- T1/2: 3–4 days
- Less adverse effects.

Pyronaridine

- MOA similar to chloroquine.
- available in fixed-dose combination therapy (3:1 ratio) combination with artesunate (Pyramax®)
- excellent potency against *P. falciparum* & chloroquine-resistant strains *P. vivax*
- T1/2: 9-13 days
- Toxicity is less than chloroquine.



แนวทางเวชปฏิบัติ

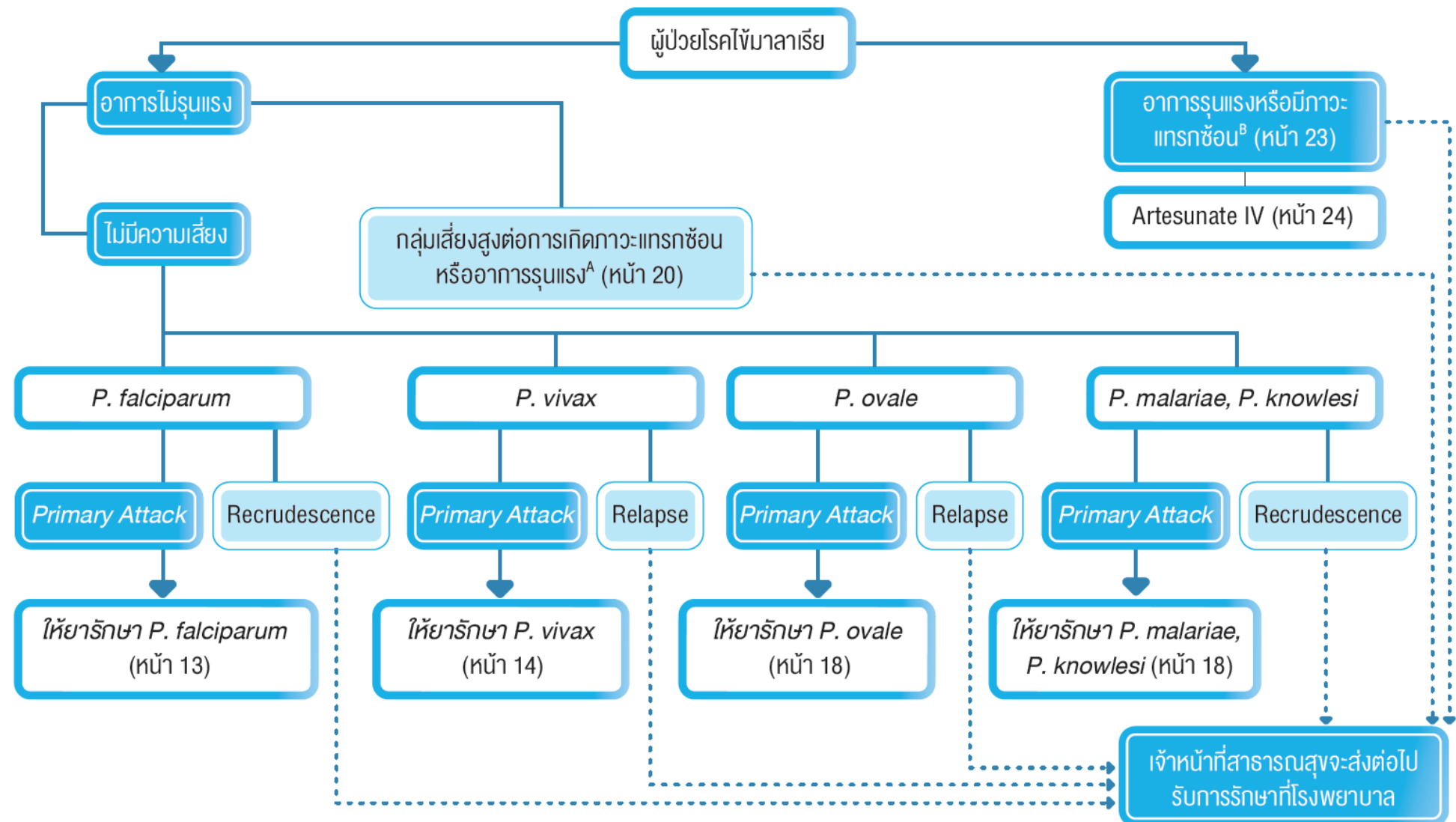
ในการรักษาผู้ป่วยโรคไข้มาลาเรีย สำหรับแพทย์
ประเทศไทย พ.ศ. 2568
โดย คณะกรรมการนโยบายและแนวทางการใช้ยารักษาโรคไข้มาลาเรีย



แผนผังที่

1

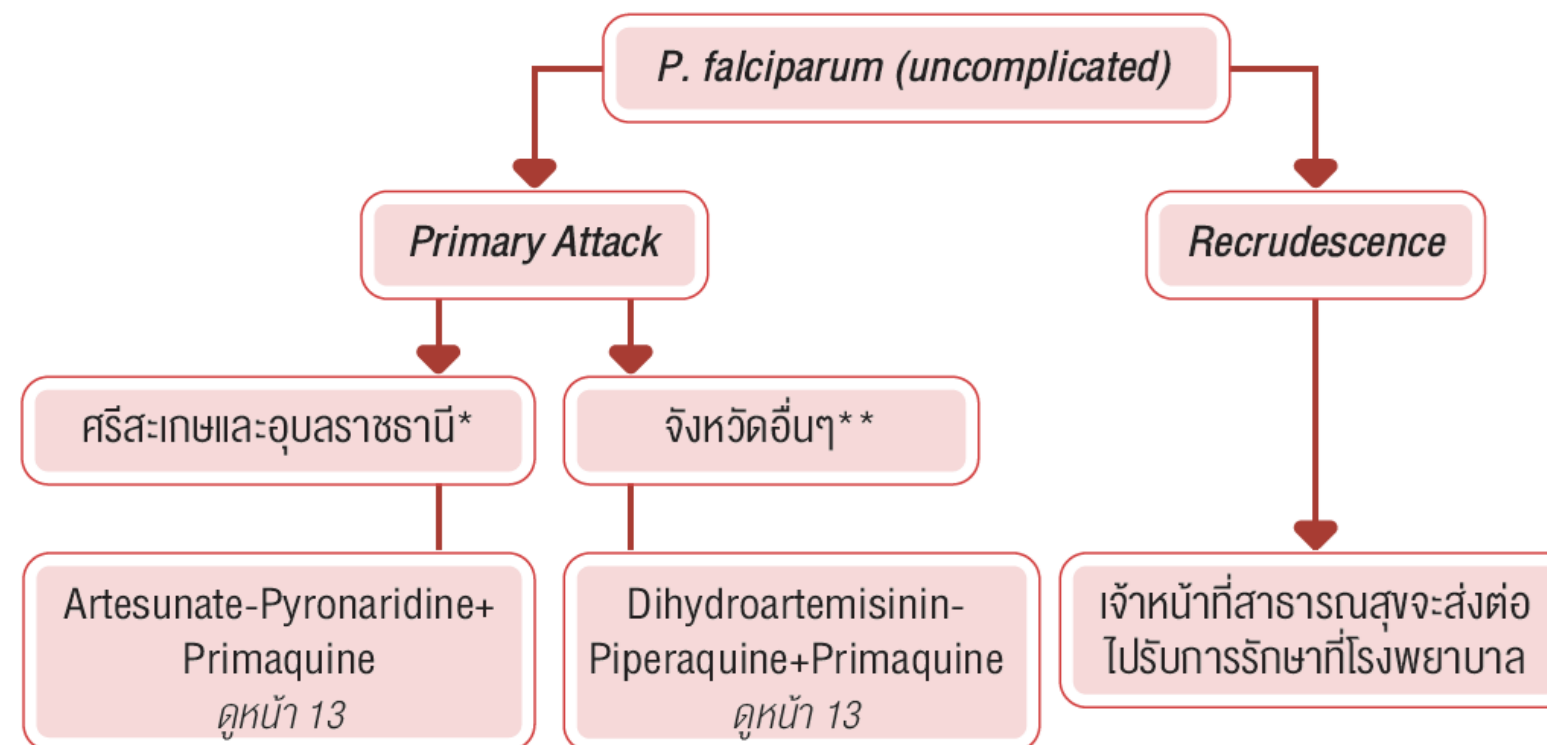
สรุปหลักการใช้ยารักษาโรคไข้มาลาเรียตามนโยบายการวินิจฉัยและดูแลรักษาโรคไข้มาลาเรียภายใต้ยุทธศาสตร์
การกำจัดโรคไข้มาลาเรีย ประเทศไทย พ.ศ. 2560 – 2569



แผนผังที่

2

→ การให้ยารักษาผู้ป่วย *P. falciparum* ที่ไม่มีภาวะแทรกซ้อน



- Artesunate-Pyronaridine (fixed dose combination) 3 days + primaquine 0.5 mg/kg/day 1 day
 - Artesunate 60 mg + Pyronaridine 180 mg
- DHA-PIP (fixed dose combination) 3 days + primaquine 0.5 mg/kg/day 1 day
 - Dihydroartemisinin 40 mg + Piperaquine 320 mg

แผนผังที่ 3 → การรักษาโรคไข้มาลาเรียชนิด *P. vivax* ที่ไม่มีภาวะแทรกซ้อน



^{*A} กลุ่มเสี่ยงสูงต่อการเกิดภาวะแทรกซ้อนหรืออาการรุนแรง

^{**} ข้อบ่งชี้และข้อห้ามใช้ยา Tafenoquine หน้า 15

ตารางที่ 3 → ขนาดของยา Chloroquine

รูปแบบยา: Chloroquine 250 mg ต่อเม็ด หรือเท่ากับ Chloroquine 150 mg base เม็ด

น้ำหนัก (kg)	วันที่ 0 (เม็ด)	วันที่ 1 (เม็ด)	วันที่ 2 (เม็ด)	รวมยาที่จ่าย (เม็ด)
น้ำหนักน้อยกว่า 11 kg	1	½	½	2
11 ถึงน้อยกว่า 15 kg	1 ½	½	½	2 ½
15 ถึงน้อยกว่า 25 kg	2	1	1	4
25 ถึงน้อยกว่า 50 kg	3	1 ½	1 ½	6
น้ำหนัก 50 kg ขึ้นไป	4	4	2	10

จะเข้าป่าต้องกินยาป้องกันมาลาเรียไหม ?

“ไม่แนะนำให้รับประทานยาป้องกันมาลาเรียในประเทศไทย ”

1. ไม่มียาใดที่ได้ผล 100% ในการป้องกัน
2. ต้องกินยานาน เสี่ยงต่อการเกิดอาการข้างเคียง
3. การกินยาป้องกันมาลาเรียอาจทำให้อาการของโรคมาลาเรียไม่ชัดเจน
4. ความชุกของโรคมาลาเรียในไทยน้อยมาก (1 ใน 16,000 คน)
5. การป้องกันไม่ให้ยุงกัด (mechanical protective measures)
 - การใช้มุ้ง (ชุบสาร pyrethrin)
 - การสวมเสื้อป้องกันร่างกายให้มิดชิด
 - การใช้ยาทากันยุงกัด: สาร n, n dimethyl-m-toluamide (DEET) 35% กันยุงนาน 4-6 ชม.



Standby drug

(Standby Emergency Treatment for malaria)

1. Malarone (Malanil[®]) 250 mg atovaquone/100 mg proguanil hydrochloride: กินวันละ 4 เม็ด ติดต่อกัน 3 วัน

2. Artesunate plus mefloquine

- เป็นยาเพื่อใช้รักษามาลาเรีย ไม่ใช่เพื่อป้องกันมาลาเรีย
- แพทย์อาจสั่งยานี้ให้นักท่องเที่ยวบางกลุ่มเพื่อให้พกติดตัวไว้ เพื่อไว้ใช้ในยามจำเป็น
 - การท่องเที่ยวในประเทศไทยไม่จำเป็นต้องพกยารักษามาลาเรีย

พิจารณากินเมื่อ

1. เข้าไปอยู่ในพื้นที่เสี่ยงต่อการติดมาลาเรียมากกว่า 1 สัปดาห์ขึ้นไป
2. มีอาการไข้ ซึ่งมักเป็นไข้สูงหนาวสั่น คลื่นไส้ อาเจียน ปวดศีรษะ ปวดตามตัว
3. ไม่สามารถพบแพทย์เพื่อตรวจรักษาได้ภายใน 12-24 ชั่วโมง
4. กินยาตามที่แพทย์สั่งเท่านั้น
5. แม้ว่าจะกินยาแล้วหรือแม้ว่าอาการดีขึ้นแล้ว ต้องไปพบแพทย์เพื่อตรวจเลือดยืนยันเสมอ

Chemoprophylaxis & Treatment During Pregnancy & Lactation

- In general: should not use any antimalarial drugs during pregnancy because lack of sufficient safety evidence.

***P. falciparum* infection**

- 1st trimester: Quinine + Clindamycin
- 2nd and 3rd trimester: Dihydroartemisinin + Piperaquine

Uncomplicated malaria:

- Chloroquine (chloroquine-sensitive parasites area)
- Quinine (*P. vivax* infections) or in combination with clindamycin (*P. falciparum* infections), in areas with chloroquine-resistant area

***P. vivax* or *P. ovale* infections:** chloroquine chemoprophylaxis until deliver -> Primaquine (after confirm that not G6PD-deficient)

Lactating mothers: most compounds are acceptable (prefer chloroquine and hydroxychloroquine)

- Primaquine: check G6PD level
- Atovaquone-proguanil: Bodyweight >5 kg

Malaria Vaccine

RTS,S/AS01 (RTS,S) (Mosquirix®)

- World's first malaria vaccine (2021)
- targets one molecule in the *P. falciparum*, circumsporozoite protein (CSP)
- 30-40% efficacy against clinical malaria in children living in regions with moderate-to-high *P. falciparum* malaria transmission.
- reduces severe malaria cases by ~30%
- Dosage & schedule: first 3 doses: at 5, 6, & 7 months of age, Booster: at 18 months
- Limitations: moderate efficacy, short-lived immunity (booster needed)

R21/Matrix-M

- Approved in 2023 Similar to RTS,S but with a higher CSP antigen dose and Matrix-M adjuvant
- 75% efficacy (higher than RTS,S)
- Dosage & schedule: 3-dose regimen + booster at 12 months



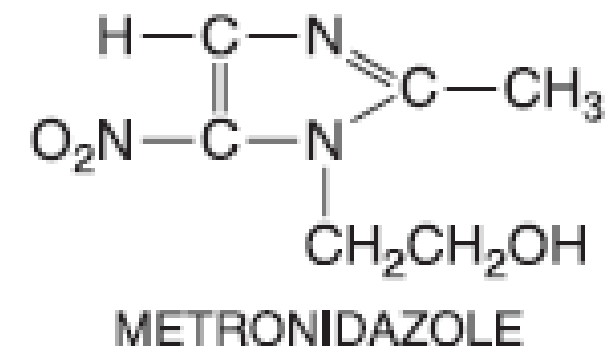
Drugs for other protozoa

Common protozoal infections & drug treatments

Organism	Disease	Common drug treatments
Amoeba		
<i>Entamoeba histolytica</i>	Amoebic dysentery	Metronidazole, tinidazole, diloxanide
Flagellates		
<i>Trypanosoma brucei rhodesiense</i> <i>Trypanosoma brucei gambiense</i>	Sleeping sickness	Suramin, pentamidine, melarpassol, eflornithine, nifurtimox
<i>Trypanosoma cruzi</i>	Chagas' disease	Nifurtimox, benzindazole
<i>Leishmania tropica</i> <i>Leishmania donovani</i> <i>Leishmania Mexicana</i> <i>Leishmania braziliensis</i>	'Kala-azar' 'Chiclero's ulcer' 'Espundia' 'Oriental sore'	Sodium stibogluconate, amphotericin, pentamidine isetionate
<i>Trichomonas vaginalis</i>	Vaginitis	Metronidazole, tinidazole
<i>Giardia lamblia</i>	Diarrhoea, steatorrhoea	Metronidazole, tinidazole, mepacrine
Sporozoa		
<i>Plasmodium falciparum</i> ^a <i>Plasmodium vivax</i> <i>Plasmodium ovale</i> <i>Plasmodium malarariae</i>	Malignant tertian malaria Benign tertian malaria Benign tertian malaria Quartan malaria	Artemether, atovaquone, chloroquine, clindamycin, dapsone, doxycycline, lumefantrine, mefloquine, primaquine, proguanil, pyrimethamine, quinine, sulfadoxine
<i>Toxoplasma gondii</i>	Encephalitis, congenital malformations, eye disease	Pyrimethamine-sulfadiazine
Ciliates and others		
<i>Pneumocystis carinii</i> ^b	Pneumonia	Co-trimoxazole, atovaquone, pentamidine isetionate

1. Metronidazole

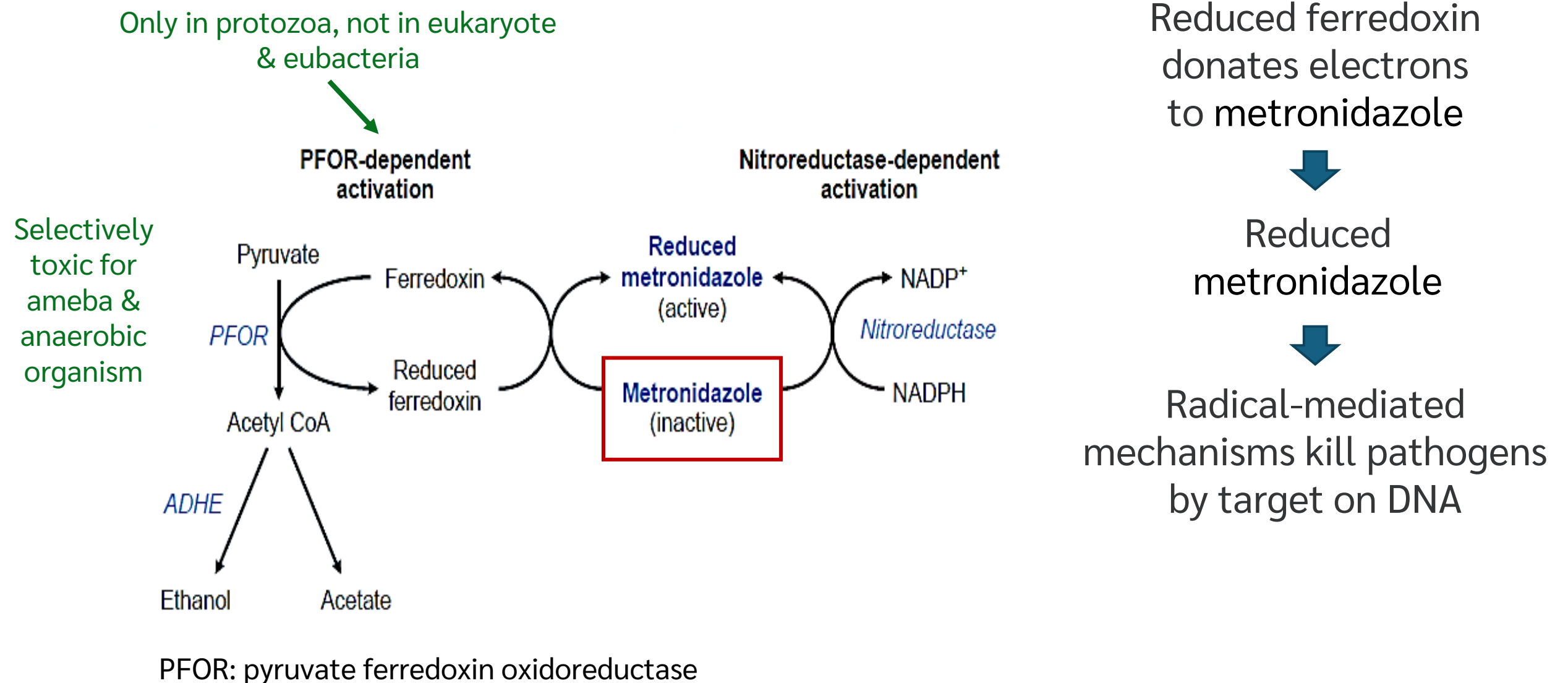
- Active against a wide variety of anaerobic protozoal parasites and anaerobic bacteria.
 - amoebiasis (*E. histolytica*),
 - *Giardia lamblia*,
 - *Trichomonas vaginalis*,
 - Anaerobic cocci,
 - Anaerobic gram-negative bacilli,
 - *Pseudomembranous colitis* in brain abscesses
 - Microaerophilic bacteria such as *Helicobacter* & *Campylobacter spp.*
- Mixed amebicides: Luminal & extra-luminal amebicide
 - Luminal (intestinal) infection – asymptomatic carriers or mild diarrhea.
 - Extra-luminal (invasive) infection – severe colitis, liver abscess, or systemic spread



1. Metronidazole

MOA:

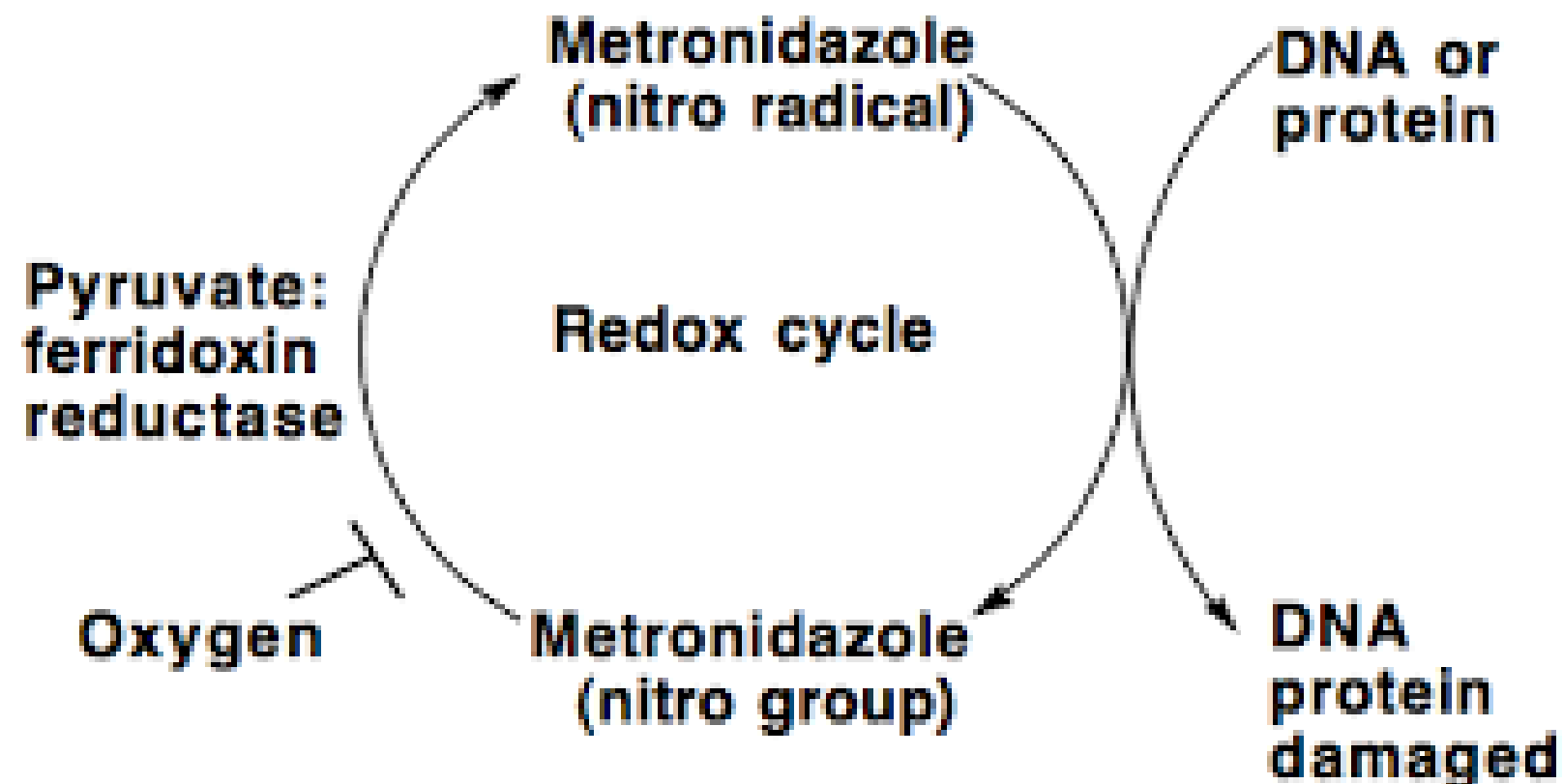
- The nitro group of metronidazole can serve as an electron acceptor, forming reduced cytotoxic compounds that bind to proteins & DNA, resulting in cell death.



1. Metronidazole

Mechanism of resistance:

- Impaired oxygen-scavenging capabilities -> increase levels of O_2 inhibit metronidazole-induced cytotoxicity (O_2 competes with metronidazole for electrons) -> decreased activation of metronidazole
- Lowered levels of PFOR and ferredoxin



1. Metronidazole

Pharmacokinetics:

- Rapidly & completely absorbed (oral, intravenous, intravaginal & topical preparations)
- Rapidly distribute tissues (include cerebrospinal fluid), except placenta
- $t_{1/2}$: 8 hrs.
- Metabolized in liver
- Excreted in urine

Adverse effects:

- Commons: headache, vomiting, nausea, dry mouth, metallic taste
- CNS symptoms: dizziness, headache, sensory neuropathies
- Stevens-Johnson syndrome (rare)
- Do not use in the first trimester of pregnancy
- Precipitate CNS signs of lithium toxicity
- Prolong the prothrombin time of patients receiving therapy with warfarin

1. Metronidazole

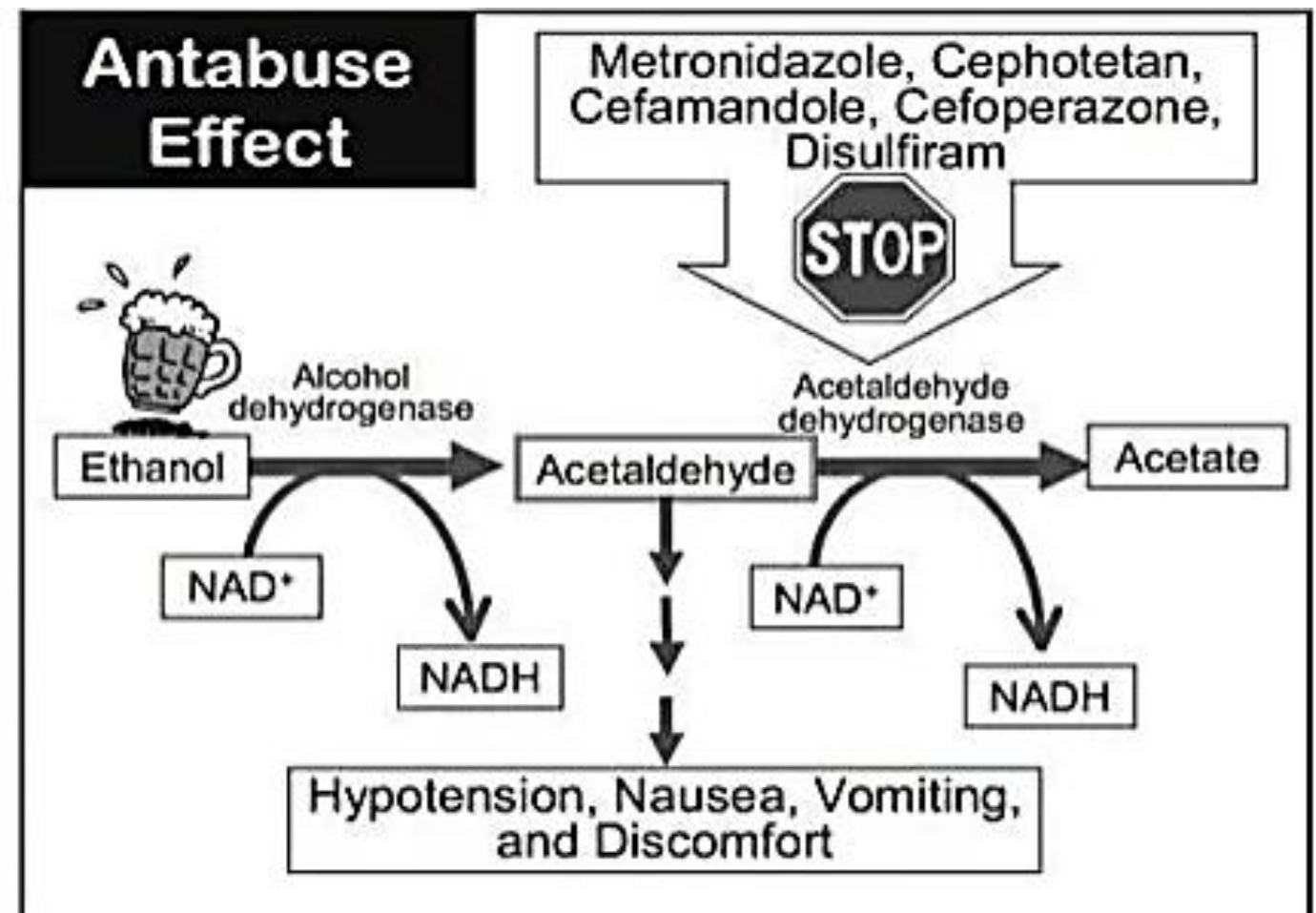
- What happens when you mix metronidazole and alcohol?
- “Disulfiram-like reaction”
 - Do not take with alcohol

Metronidazole & Alcohol

www.medinaz.com

It can cause **disulfiram** like reaction (e.g. flushing, nausea, vomiting, palpitations, tachycardia, psychosis)

No alcohol products during metronidazole therapy



1. Metronidazole

Indication	Standard dosage (adult)
Bacterial infections	15 mg/kg intravenous infusion over 1 hour, then 7.5 mg/kg administered intravenously every 6 hours for 7-10 days or 7.5 mg/kg taken orally every 6 hours for 7-10 days
Pelvic inflammatory disease (PID)	500 mg taken orally every twice daily for 14 days
Bacterial vaginosis	500 mg twice daily for 7 days or one applicatorful (5 grams of 0.75% gel) administered intravaginally once daily for 5 days
Trichomoniasis	2 g taken as a single dose or divided into two 1-g doses or 500 mg twice daily for 7 days (dosage for HIV-infected women) or one 250 mg tablet taken orally 3 times daily for 7 days or one 375 mg capsule taken twice daily for 7 days
Acute intestinal amebiasis	750 mg taken orally 3 times daily for 5-10 days
Amebic liver abscess	500 or 750 mg taken orally 3 times daily for 5-10 days
Infection prevention in patients undergoing colorectal surgery	500 mg or 15 mg/kg administered IV and completed within 1 hour before surgery, then 7.5 mg/kg IV every 6 hours for 2 doses or 1 g taken orally 3 times over 10 hours starting the afternoon and evening before the day

2. Tinidazole

- a second-generation nitroimidazole
- Higher cure rate for protozoal infections & a better toxicity profile than metronidazole
- Similar to metronidazole in the spectrum of activity, absorption, adverse effects and drug interactions.
- As effective as metronidazole, with a shorter course of treatment.

Metronidazole vs. Tinidazole
ต่างกันอย่างไร ยาตัวไหนดีกว่ากัน?



	Metronidazole (M)	Tinidazole (T)
Drug class	Nitroimidazole antimicrobial	
Available formulations	Oral, topical (cream/gel), intravaginal, intravenous	Oral
Indications	• Amebiasis: M 750 mg x 3 x7 days, T 2 g x1 x 3 days • Trichomoniasis (tinidazole is useful in treating trichomoniasis that fails to respond to metronidazole): M 2 g once or 250 mg x3 x 7 days, T 2 g once • Bacterial vaginosis	
	Anaerobic Bacterial Infections: • Intra-abdominal Infections (peritonitis, intra-abdominal abscess, liver abscess) • Skin and skin structure infections • Endometritis, endomyometritis, tubo-ovarian abscess, postsurgical vaginal infections • Septicemia • Meningitis, brain abscess • Pneumonia, empyema, lung abscess • Endocarditis	Giardiasis (diarrhea) Tinidazole is more effective than metronidazole for giardiasis: 2 g once
Half-life	8 hrs.	10-15 hrs
Food	Food does not affect the oral bioavailability. Both metronidazole & tinidazole can be taken with food to minimize gastrointestinal side effects	

Drugs	Therapeutic Uses	Clinical Pharmacology and Tips
Amebiasis		
Metronidazole	•Amoebic colitis and liver abscess	•Always administer with luminal agent •Orally administered: > 80% bioavailable •Common side effects: headache and metallic taste •Can have disulfiram-like effect
Tinidazole	•Amoebic colitis and liver abscess	•Always administer with luminal agent •<u>Can have disulfiram-like effect</u>
Giardiasis		
Metronidazole	•Giardiasis	•5-day course •Not FDA-approved for indication, but years of experience
Tinidazole	•Giardiasis	•Single dose sufficient
Trichomoniasis		
Metronidazole	•Trichomoniasis	•Drug of choice •2 g once •If failure, give second dose in 4–6 weeks
Tinidazole	•Trichomoniasis	•2 g once •Can be used for resistant infection

3. Pentamidine

- **Broad-spectrum against protozoa and fungi** (generally reserve for severe infection)
 - Pneumocystis Pneumonia (PCP) caused by *Pneumocystis jirovecii*
 - Cutaneous Leishmaniasis caused by *Leishmania* species
 - African Trypanosomiasis caused by *Trypanosoma brucei gambiense* (early stage)
- MOA: inhibiting DNA, RNA, phospholipid & protein synthesis
- Highly charged compound is poorly absorbed orally.
- IV, IM injections, Inhalation

Adverse effects: quite toxic!

- Serious toxicities: kidney damage, hepatic impairment, blood dyscrasia, hypoglycemia, hypocalcemia, hypomagnesemia, hyperkalemia
- Others: Hypotension, tachycardia, and headache (prevented by slowing the infusion rate)

3. Pentamidine

Pentamidine					
Pharmacokinetic Properties					
	Value	Units	Prep. and Route of Admin.	Reference	Comments
Absorption					
Bioavailability					Poor bioavailability when given orally
Distribution					
Volume of Distribution	140	l	i.v.	Conte et al (1986)	
Plasma Protein Binding	70	%		Abdi et al (1995)	
Metabolism					
Plasma Half-Life	0.9	h	i.v.	Conte et al (1986)	
Bio Half-Life					
Clearance					
Routes of Elimination					

4. Anti-toxoplasma agents

Drugs	Mechanism of action	Clinical Pharmacology	ADRs
Pyrimethamine	• dihydrofolate inhibitor • inhibit DNA & protein synthesis	• Combine sulfadiazine or clindamycin • Give with leucovorin	bone marrow suppression
Sulfadiazine	• inhibitor of dihydropteroate synthetase, which is essential in the folate synthesis pathway	• Combine with pyrimethamine and folic acid	bone marrow suppression
Clindamycin	• Acute toxoplasmosis	• Combine with pyrimethamine	Use if cannot tolerate sulfonamide



Anthelmintic drugs



Classification of anthelmintic drugs

- **Drugs affecting microtubule function**
 - Benzimidazoles: albendazole, mebendazole
- **Drugs causing neuromuscular paralysis**
 - Ivermectin, Pyrantel pamoate, Levamisole, Piperazine
 - Praziquantel
- **Drugs affecting energy metabolism**
 - Praziquantel, Niclosamide
- **Other anthelmintic agents**
 - Diethylcarbamazine (DEC)

Types of helminths

Nematodes (roundworms) – พยาธิตัวกลม

พยาธิไส้เดือนกลม (*Ascaris lumbricoides*, Roundworm)

พยาธิเส้นด้าย (*Strongyloides stercoralis*, Threadworm)

พยาธิปากขอ (*Necator americanus*, Hookworm)

พยาธิตัวจิ๊ด (*Gnathostoma spinigerum*)

พยาธิแส้ม้า (*Trichuris trichiuria*, whipworm)

- Benzimidazoles:
Mebendazole,
Albendazole
- Ivermectin
- Pyrantel pamoate

Cestodes (tapeworms) – พยาธิตัวแบน

พยาธิตืดหมู (*Taenia solium*),

พยาธิตืดวัว (*Taenia saginata*)

- Praziquantel
- Niclosamide

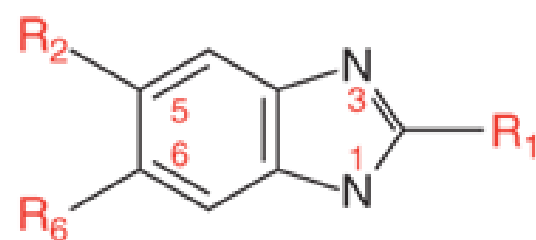
Trematodes (flukes) – พยาธิใบไม้

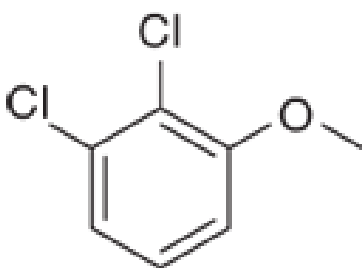
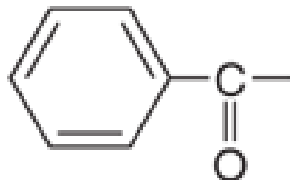
พยาธิใบไม้ในเลือด (*Schistosoma* species, blood fluke),

พยาธิใบไม้ในตับ (*Opisthorchis viverrine*, liver fluke)

- Praziquantel
- Triclabendazole

Benzimidazoles: Albendazole, Mebendazole



R_2	R_6	R_1	DERIVATIVE
	Cl	S-CH ₃	Triclabendazole
	H	—NHCO ₂ CH ₃	Mebendazole
	H	—NHCO ₂ CH ₃	Albendazole

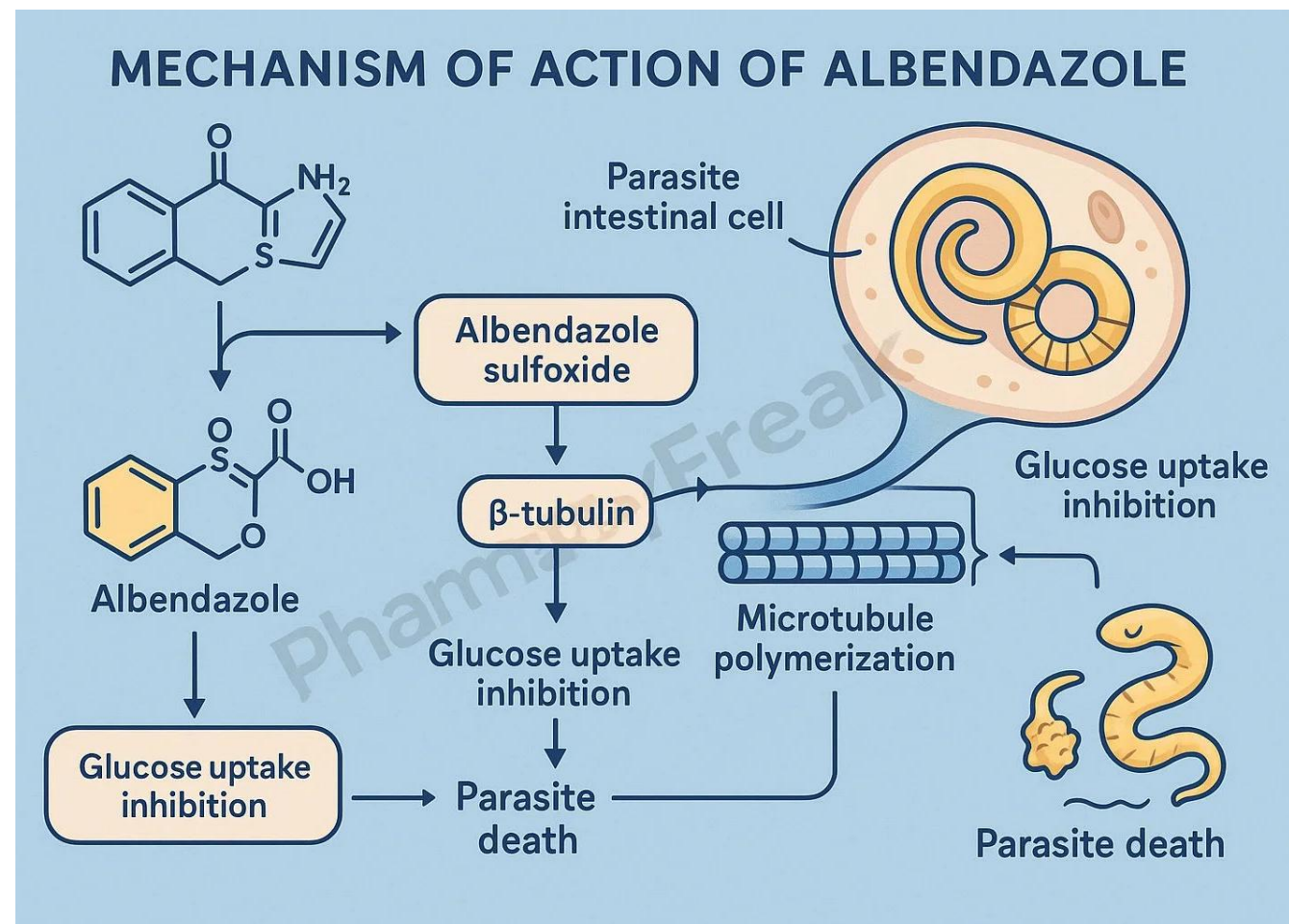
- Broad-spectrum anthelmintic
- Albendazole & Mebendazole – widely used
- Triclabendazole – reserve for liver fluke (พยาธิใบไม้ตับ) & paragonimiasis (พยาธิใบไม้ปอด)

Source: Laurence L. Brunton, Björn C. Knollmann:
Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e:
Copyright © McGraw Hill. All rights reserved.

Benzimidazoles: Albendazole, Mebendazole

MOA

binds to the colchicine-sensitive
site of β -tubulin subunit
↓
inhibits microtubule
polymerization
↓
inhibit microtubule-dependent
glucose uptake
↓
inhibits glycogen storage
↓
parasite immobilization & death



- 250–400 times more selective inhibitory in helminth than in mammalian tissue.
- Cure rates 60 - 100% (depend on helminths)



Pharmacokinetics

Albendazole	Mebendazole
• A: poorly absorbed in GI -> good for intestinal helminth infection • Fatty food: ↑ absorption 5X (better than mebendazole) • D: Widely distributed throughout the body. (protein binding 70%) • M: metabolized by CYP3A4 active metabolites: albendazole sulfoxide (more active than parent) • E: urine and bile • Half-life: 8-12 hrs.	• A: low systemic bioavailability (22%) • M: First pass metabolism by CYP3A4 • rapidly metabolized to an inactive compound • E: mainly feces • Half-life: 2-6 hrs.

Therapeutic uses

Albendazole

- Highly effective for GI nematode infections
- is preferred for systemic infections like neurocysticercosis and hydatid disease.

Doses:

- Adults & children > 10 kg: 400 mg dose (2 × 200 mg tablets)
- Children > 6 months and < 10 kg: 200 mg
- พยาธิเข็มหมุด ไล่เดือน ปากขอ: 400 mg ครั้งเดียว
- พยาธิแส้ม้า: 400 mg x 3 days
- พยาธิตัวจิ๋ว: 400 mg x 2 x 21 days
- Neurocysticercosis: 5 mg/kg x 3 x 1 month
- Larval forms of Taenia solium: 400 mg x 2 x 8–30 days

Mebendazole

- GI nematode infections

Doses:

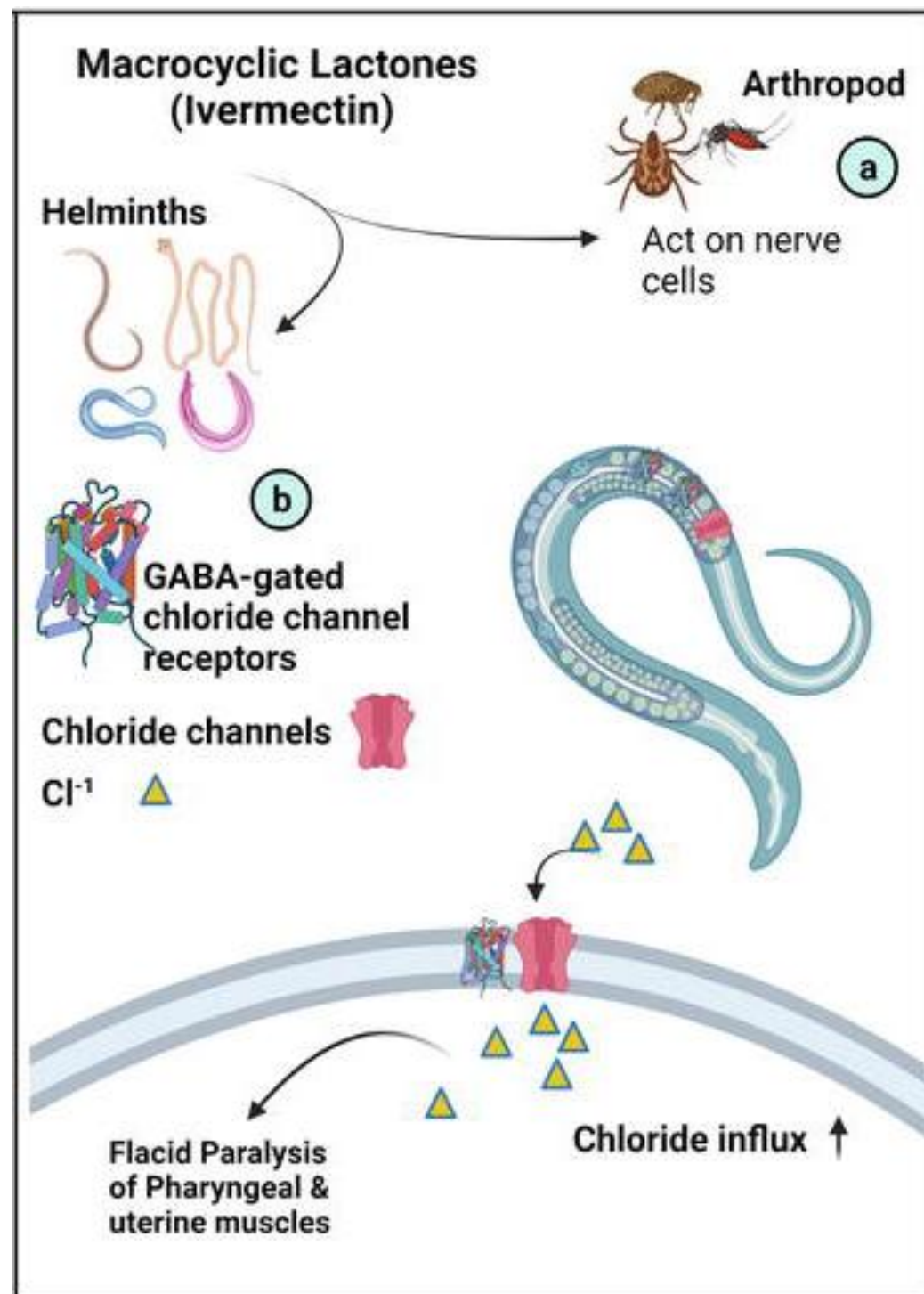
- พยาธิเข็มหมุด: 100 mg ครั้งเดียว (ซ้ำ 2 weeks)
- พยาธิไล่เดือน แส้ม้า ปากขอ: 100 mg x 2 x 3 days หรือ 500 mg ครั้งเดียว



Adverse Effects

Albendazole	Mebendazole
very good safety profiles when administered in short courses	
• headache, anorexia, nausea, vomiting, dizziness	• anorexia, nausea, vomiting, dizziness, diarrhea, headache, elevated liver enzyme
• long-term use: liver dysfunction (elevated liver enzyme) • myelosuppression • liver function tests and CBC	

Ivermectin



MOA:

- Activating glutamate-gated chloride channels, leading to increased chloride influx → hyperpolarization of the nerve or muscle cell → paralysis → parasite death.
- Also bind with GABA receptors in nematodes & insects but does not cross the blood-brain barrier in humans.



Ivermectin

Therapeutic uses:

- Filarial infections
 - 400 µg/kg single annual dose
 - combination: ivermectin (200 µg/kg) + albendazole (400 mg) - single annual dose (more effective)
- Onchocerciasis
 - 150-200 ug/kg single dose, given every 6-12 months
 - causes “river blindness”
- Strongyloidiasis - 150-200 ug/kg single dose
- Cutaneous larva migrans – 200 ug/kg
- Others: common roundworms, whipworms, threadworms but not hookworms.



Ivermectin

PK:

- Absorption: oral bioavailability increases with fatty meals
- Distribution: distributed widely throughout the body
- Metabolism: Liver via CYP3A4 (major), CYP2D6 (minor), CYP2E1 (minor)
- Excretion: mainly feces, minimal urine (<1%)
- Half-life: 18-36 hrs.

Adverse effects:

- Common: Nausea, dizziness, diarrhea, rash, mild itching
- Severe (especially in onchocerciasis):
 - Mazzotti reaction (fever, headache, rash, lymphadenopathy) due to rapid parasite death.
 - Neurological toxicity (rare in humans but caution in patients with brain disorders, more common in dogs).



Pyrantel pamoate

MOA:

- induce persistent activation of nicotinic acetylcholine (nACh) receptors -> spastic paralysis of the worm

Therapeutic uses:

- Treatment of pinworm, roundworm & hookworm infections
- Dose: 1 mg/kg (maximum 1 g) OD for 3 days

Adverse effects:

- Usually well tolerated



Levamisole

MOA:

- potent agonist of the L-type nematode nicotinic acetylcholine receptor -> paralyze worms -> expelled in the faces
- inhibit fumarate reductase (enzyme required for succinate production – deplete ATP production -> parasite death)

Therapeutic uses:

- Treatment of common roundworm, less effective in threadworm
- Adult & Pediatric (≥ 6 years): 2.5 mg/kg as a single dose, maximum dose 150 mg/kg (Repeat dose after 7 days if needed.)



Levamisole

PK:

- quickly absorbed from the GI tract
- mainly metabolized in the liver
- Half-life: 4 hrs.

Adverse effects:

- Nausea, vomiting, abdominal pain, Dizziness, headache, Metallic taste
- Rare: hematologic toxicity (agranulocytosis) – limit its clinical use.



Piperazine

MOA:

- inhibits acetylcholine at the neuromuscular junction → paralyzes helminths
- act as GABA receptor agonist → increase chloride ion influx → hyperpolarization at muscle fibers → paralysis of nematodes →
- live worms are expelled by peristalsis

Therapeutic uses:

- an alternative for the treatment of roundworm (not other helminths)
- 5 mg/kg (maximum dose, 3.5 g) orally once daily for 2 days



Piperazine

Adverse effects:

- nausea, vomiting, diarrhea, abdominal pain, dizziness, headache
- Rare: neurotoxicity
- Should not be given to pregnant women, patients with impaired renal or hepatic function, or those with a history of epilepsy or chronic neurologic disease.



Praziquantel

MOA:

- binding to protein kinase C-binding sites in β - subunit of schistosome voltage-gated calcium channels



disrupts Ca^{2+} homeostasis in the parasite



paralysis and death of the worm

- disrupts the tegument of the parasite (unmasking antigens) → more susceptible to host's immune responses
- inhibit adenosine uptake -> energy depletion



Praziquantel

PK:

- A: well absorbed after oral administration
- D: can distribute into the CSF
- M: rapidly metabolized to inactive metabolites
- E: urine
- Half-life: 60–90 hrs.

Therapeutic uses:

- Highly active against a broad spectrum of trematodes and cestodes.
- main treatment for schistosomiasis (blood fluke infection)
- safe for pregnant and lactating women



Praziquantel

Adverse effects:

- abdominal discomfort
- dizziness
- headache



Niclosamide

MOA:

- Blocks oxidative phosphorylation & ATP production in mitochondria.
- Disrupts glucose uptake in tapeworms -> starvation & death
- No direct paralysis effect.

Therapeutic uses:

- Effective against tapeworm infections
- Not effective against nematodes or flukes

Adverse effects:

- nausea, vomiting, pruritus, light-headedness



Diethylcarbamazine (DEC)

MOA:

- inhibit parasite cholinergic muscle receptors
- disrupt of microtubule formation
- alter helminthic surface membranes -> enhanced killing by the host's immune system

Therapeutic uses:

- Mainly effective against filarial parasites
- lymphatic filariasis (โรคเท้าช้าง) and loiasis (African eye worm)
- 6 mg/kg for 6-12 days
- Often combined with albendazole or ivermectin in mass drug administration (MDA) programs.



Diethylcarbamazine (DEC)

PK:

- Well absorbed after oral administration
- rapidly metabolized to inactive metabolites
- Excretion: urine (~50% as unchanged drug; ~10% as metabolite); feces (~10%)
- Half-life: 10-12 hrs.

Adverse effects:

- GI disturbances, joint pain, headache, weakness
- Allergic side effects because of host response to dying filarial
- prevent by pretreatment with glucocorticoids & antihistamine
- Safe for use during pregnancy



Summary

- **Spectrum of activity:** Some drugs (e.g., albendazole, praziquantel) have broad-spectrum activity, while others (e.g., ivermectin) target specific parasites.
- **Resistance:** Emerging resistance, particularly in filarial and soil-transmitted helminths, is a growing concern.
- **Adverse effects:** Most anthelmintics are well tolerated, but side effects may include GI disturbances, dizziness, headache, or hypersensitivity reactions.
- **Drug interactions:** Some drugs, such as praziquantel and albendazole, may have interactions with CYP enzyme inducers, affecting their efficacy.



Mahidol University
Faculty of Pharmacy

Thank you